Phase mixing against a passive-scalar cascade: the Hermite spectrum of kinetic reduced magnetohydrodynamic turbulence

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Abstract

In kinetic reduced magnetohydrodynamics (KRMHD) the compressive fluctuations are a kinetic field carried by the Alfvénic turbulence. Phase mixing moves their free energy to fine structure in parallel velocity and the turbulence moves it to small perpendicular scales, and the Hermite spectrum $W(m)$ records the competition between the two. We show that the compressive sector is exactly a continuum of passive scalars, one for each parallel phase velocity, each carried by the $\boldsymbol{E} \times \boldsymbol{B}$ flow and streaming along the perturbed magnetic field at its own speed. The nonlinearity therefore moves no free energy between phase velocities and, while the field lines stay frozen into the flow, none between field-aligned wavenumbers. We propose that its only effect on $W(m)$ is to remove free energy from every Hermite moment at the same rate, the rate at which the turbulence carries a passive scalar to the dissipation scale, while phase mixing proceeds at the field-aligned wavenumber at which the free energy was injected. This gives $W(m) \propto m^{-1/2}$ at small m and a decay exponential in $m^{1/2}$ beyond a break at $m_* \sim \beta_i/\chi^2$, where χ is the nonlinearity parameter. In simulations of forced KRMHD turbulence at $\beta_i = 1$, the model, with its two inputs measured independently, reproduces $W(m)$ and the Hermite flux along the field lines, and there is no return flux from high to low m. Lowering β_i moves the break into the resolved range, as predicted. With the Hermite resolution raised until the break is resolved, m_* grows about in proportion to β_i for $0.05 \leq \beta_i \leq 0.3$. Against χ, at $\beta_i = 0.05$, it grows faster than χ^{-2}, and about as χ^{-4} between $\chi \approx 0.56$ and 0.40, where the perpendicular sink per outer eddy time weakens as the turbulence becomes weak. The model is too simple in one respect: the perpendicular sink is not the same at every m but grows with the time since the free energy was injected, in the same way at every β_i, so at $\beta_i = 0.3$ and 0.1 the spectrum lies above the model. At $\beta_i = 0.3$ the field-aligned spectrum of the free energy does not change with the resolution, apart from a tail holding less than 0.5% of it, but the time in which the sink removes the free energy has not converged: it grows by about the same amount with each doubling up to 256^3 grid points, so whether the break becomes independent of the dissipation scale, as the model assumes, remains open.

1 Introduction

Kinetic reduced magnetohydrodynamics (KRMHD), the limit $k_\perp \rho_i \to 0$ of gyrokinetics derived by Schekochihin et al. (2009), hereafter S09,[1] describes anisotropic fluctuations at scales much larger than the ion gyroradius ρ_i in a strongly magnetised, weakly collisional plasma. Its Alfvénic fluctuations obey reduced magnetohydrodynamics (RMHD). Its compressive fluctuations, the slow waves and the entropy mode, obey a kinetic equation for the ion distribution in the parallel velocity

[1] Appendix E compares our notation with that of S09 and of Chandran et al. (2026).

$v_\parallel$; they are carried by the Alfvénic fluctuations and do not act back on them. Without collisions they are damped by phase mixing (Barnes, 1966; S09, §6.2), which moves their free energy to ever finer structure in $v_\parallel$, that is, to large Hermite order m in a Hermite expansion in $v_\parallel$ (Zocco and Schekochihin, 2011; Kanekar et al., 2015; Schekochihin et al., 2016). In turbulence the same free energy is also carried to small perpendicular scales. Which of the two routes it takes decides whether the compressive cascade heats ions through Landau damping or reaches the ion gyroscale (S09; Schekochihin et al., 2016; Meyrand et al., 2019; Kawazura et al., 2019, 2020), and the Hermite spectrum $W(m)$ is the most direct record of that competition.

Two pictures of $W(m)$ in turbulence have been proposed. In the first, nonlinear advection couples fluctuations with parallel wavenumbers of opposite sign and so returns free energy from high to low m, as a stochastic plasma echo (Kanekar, 2015; Schekochihin et al., 2016; Parker et al., 2016; Adkins and Schekochihin, 2018). Meyrand et al. (2019) found in KRMHD turbulence at $\beta_i = 1$ a spectrum somewhat steeper than m^{-1}, against $m^{-1/2}$ for linear phase mixing, and a net flux to high m much smaller than the injection rate away from the forcing scales, and attributed both to such echoes. In gyrokinetic simulations Kawazura et al. (2019, 2020) found $W(m) \propto m^{-1/2}$ at high β_i and close to m^{-1} at $\beta_i = 0.1$, and read the steeper spectrum as phase mixing suppressed by echoes. In the second picture (S09, §6.3), the compressive fluctuations have no parallel cascade: the field-aligned wavenumber at which they were injected stays fixed while the Alfvén waves carry them to smaller perpendicular scales, so at small enough scales the perpendicular cascade outruns the damping. The two pictures differ on whether the compressive fluctuations acquire structure along the moving field lines. Meyrand et al. (2019) argued that they do, because resistive reconnection relabels the field lines within about one eddy-turnover time and the compressive fluctuations then inherit the parallel cascade of the Alfvénic ones.

We make the second picture quantitative and test it. Section 3 shows that the compressive sector of KRMHD is exactly a continuum of passive scalars, or slices, labelled by the parallel phase velocity u, each carried by the $\boldsymbol{E} \times \boldsymbol{B}$ flow and streaming along the perturbed field at its own speed. The nonlinearity therefore moves no free energy between phase velocities, and free energy moves between Hermite moments only as the correlations between slices change (section 4). When the induction equation is ideal, the field lines are frozen into the flow that carries all the slices, and every volume-integrated single-point statistic, including $W(m)$, evolves as in linear phase mixing along the initial field lines (section 5). In turbulence the slices are also cascaded to small perpendicular scales and dissipated there. Section 6 proposes that this is the only effect of the nonlinearity on $W(m)$: every Hermite moment loses free energy at the same rate $1/T$, the rate at which the turbulence carries a passive scalar to the dissipation scale, while phase mixing proceeds at the field-aligned wavenumber κ of the injected free energy. The steady state is

$$W(m) = \frac{\varepsilon}{s\kappa\sqrt{2m}} \exp\Big(-\frac{\sqrt{2m}}{s\kappa T}\Big), \qquad m_* = \frac{(s\kappa T)^2}{2} \sim \frac{\beta_i}{\chi^2}, \tag{1}$$

for injection rate ε, $s = v_{\rm th}/v_A = \beta_i^{1/2}$ and nonlinearity parameter χ, summed over the distribution of κ. Two times set the spectrum: free energy injected at κ phase mixes up to moment m in the time $\sqrt{2m}/(s\kappa)$, which is shorter at higher β_i, and the perpendicular cascade removes it in the time T, which is set by the Alfvénic turbulence. The exponential in (1) is the fraction of the free energy that survives the sink while it phase mixes up to m, and m_* is the moment it reaches in the time T. Every quantity in (1) can be measured independently of $W(m)$, so the model has no free parameter. It predicts that the perpendicular dissipation of each Hermite moment is proportional to $W(m)$, that $W(m)$ and the Hermite flux computed along the field follow the model with no deficit at large m, that at fixed χ the break m_* grows with β_i, in proportion to it if T does not depend on β_i, that

T does not depend on the dissipation coefficients, and that the field-aligned spectrum of the free energy stays at the wavenumbers of the injected pattern as the resolution increases.

Section 7 tests these predictions in forced KRMHD turbulence with the GANDALF code (Kanekar, 2025). At $\beta_i = 1$ and $\chi \approx 1$ the perpendicular dissipation per Hermite moment is proportional to $W(m)$ to within 20%, and the model reproduces $W(m)$ to within 8% for $m \leq 30$ and the field-aligned flux to within 7% for $m \leq 24$. At $\beta_i = 1$ the break m_* lies beyond the $M = 32$ moments we resolve. At $\beta_i = 0.3$ and 0.1, with the same Alfvénic turbulence, it moves into the resolved range as predicted, and $W(m)$ steepens accordingly. There the measured spectrum lies above the model at every m, whereas a return flux would put it below the model at large m. The excess comes from the assumption of a single time T. The sink switches on over about five outer eddy-turnover times, as the cascade carries fresh free energy to the dissipation scale, so it removes less of the free energy at small m, which phase mixing brought there more recently. At lower β_i phase mixing is slower and the free energy at a given m is older, so the sink differs between the runs at a given m but is the same function of the time since injection in all of them. Doubling the number of Hermite moments leaves $W(m)$ unchanged to within 5% for $m \leq 10$. Raising the perpendicular resolution does not: up to 256^3 grid points T grows by about the same amount with each doubling, so the sink is not independent of the dissipation at the resolutions we reach. The other approximation of the model, a frozen field-aligned wavenumber, holds at every resolution: at $\beta_i = 0.3$ the field-aligned spectrum of the free energy does not change from 64^3 to 256^3, apart from a tail holding less than 0.5% of it. With the T measured at each resolution the model still describes $W(m)$, and there is still no return flux. With the parallel flow forced in place of the density, as by Meyrand et al. (2019), the spectrum steepens at small m, by the amount the model gives with the injected distribution of κ. With the Hermite resolution raised until the break is resolved, m_* grows as $\beta_i^{1.09\pm0.04}$ for $0.05 \leq \beta_i \leq 0.3$ at $\chi \approx 1$. At $\beta_i = 0.05$ it grows somewhat faster than χ^{-2} down to $\chi \approx 0.56$ and about as χ^{-4} between $\chi = 0.56$ and 0.40. In units of the outer eddy time the sink weakens at the smallest χ, so the estimate $m_* \sim \beta_i/\chi^2$, which takes T proportional to the eddy time, holds only near $\chi \approx 1$.

The ingredients of (1) are not new, and we credit them where they are used. The rate comparison between Landau damping at a frozen parallel wavenumber and the perpendicular cascade is S09's (§6.3, eqs 217–220). The functional form is that of Zocco and Schekochihin (2011, eq. 112) for growing modes, with their growth rate replaced by the cascade rate, and Schekochihin et al. (2016, eqs 2.58–2.59) solved the same continuity equation in m with a collisional sink. The break scale is the threshold $k_\parallel v_{\rm th} \sim \sqrt{m}\, k_\perp u_\perp$ of Schekochihin et al. (2016, eq. 4.5), and Kawazura et al. (2019) stated $m \sim \beta_i$ for critically balanced turbulence. A Hermite spectrum decaying exponentially in $\sqrt{m}$, at a rate proportional to the ratio of the cascade rate to $k_\parallel v_{\rm th}$, was seen in simulations of a kinetic passive scalar advected by a two-dimensional flow (Kanekar, 2015, chapter 4). What is new here is the exact statement that in three-dimensional KRMHD with ideal induction the nonlinearity moves no free energy in m, the model with independently measured inputs that follows from it, and its test in turbulence with diagnostics computed along the field. The slice decomposition itself has precedents in the G-transform of Morrison and Pfirsch (1992) and Morrison (2000), its application to drift kinetics by Heninger and Morrison (2018), and the per-velocity invariants of Plunk and Parker (2014) and Plunk (2015).

Section 2 sets out the compressive sector of KRMHD, sections 3 to 5 give the exact results, section 6 the model and its predictions, section 7 the simulations, and section 8 discusses what the results do and do not establish.

2 The compressive sector of KRMHD

For a plasma with one ion species and isothermal electrons, KRMHD (Schekochihin et al., 2009) consists of reduced magnetohydrodynamics (RMHD) for the Alfvénic fluctuations and a kinetic equation for the compressive ones. We use the form of CMM (Chandran et al., 2026), whose section 3.2 and appendix B restate and extend S09; their equation numbers refer to arXiv:2607.27981v1. Appendix E compares our notation with that of CMM, S09 and the GANDALF code (Kanekar, 2025).

2.1 Alfvénic fields

We set the Alfvén speed $v_A = 1$ unless stated otherwise and take the mean field $\boldsymbol{B}_0$ along $\hat{\boldsymbol{z}}$. The stream function Φ and the flux function Ψ give the $\boldsymbol{E}\times\boldsymbol{B}$ velocity $\boldsymbol{u}_E = \hat{\boldsymbol{z}}\times\nabla_\perp\Phi$ and the perpendicular magnetic fluctuation $\boldsymbol{b}_\perp \equiv \delta\boldsymbol{B}_\perp/(4\pi n_i m_i)^{1/2} = \hat{\boldsymbol{z}}\times\nabla_\perp\Psi$, and to first order the unit vector along the field is $\hat{\boldsymbol{b}} = \hat{\boldsymbol{z}} + \boldsymbol{b}_\perp$; both $\boldsymbol{u}_E$ and $\hat{\boldsymbol{b}}$ are divergence-free. With the bracket $\{f,h\} = \partial_x f\,\partial_y h - \partial_y f\,\partial_x h = \hat{\boldsymbol{z}}\cdot(\nabla f\times\nabla h)$, advection by $\boldsymbol{u}_E$ is $\{\Phi,\cdot\}$ and the derivative along the field is $\nabla_\parallel \equiv \hat{\boldsymbol{b}}\cdot\nabla = \partial_z + \{\Psi,\cdot\}$. The RMHD equations are (Schekochihin et al., 2009, eqs 17–18)

$$\partial_t\Psi + \{\Phi,\Psi\} = \partial_z\Phi, \qquad \partial_t\nabla_\perp^2\Phi + \{\Phi,\nabla_\perp^2\Phi\} = \nabla_\parallel\nabla_\perp^2\Psi. \tag{2}$$

The first, the induction equation $\partial_t\Psi = \nabla_\parallel\Phi$, states that the electric field along the total magnetic field vanishes. Equations (2) are CMM's eqs (3.23)–(3.24) in their normalised potentials (3.2) and do not contain the compressive fluctuations. The Elsasser potentials $z^\pm = \Phi \pm \Psi$ (S09's $\zeta^\pm$) obey $\partial_t z^\pm = \pm\partial_z z^\pm$ linearly, so z^+ propagates toward $-z$ and z^- toward $+z$.[2]

2.2 Kinetic equation and the two branches

The ion thermal speed is $v_{\rm th} = (2T_i/m_i)^{1/2}$, so $s \equiv v_{\rm th}/v_A = \beta_i^{1/2}$; we write $\tau = T_i/T_e$ and $Z = q_i/e$. The compressive part of the ion distribution is the function g of S09 (CMM's g_{09}, their eq. 3.26), which obeys CMM's eq. (3.25) (S09 eq. 157 without collisions),

$$\frac{\mathrm{d}g}{\mathrm{d}t} + v_\parallel\nabla_\parallel\Big[g + \Big(\frac{Z}{\tau}\frac{\delta n}{n} + \frac{v_\perp^2}{v_{\rm th}^2}\frac{\delta B}{B}\Big)F_0\Big] = 0, \qquad \frac{\mathrm{d}}{\mathrm{d}t} = \partial_t + \{\Phi,\cdot\}, \tag{3}$$

where F_0 is the three-dimensional Maxwellian. Quasineutrality and perpendicular pressure balance give (CMM eqs 3.27–3.28; S09 eqs 158–159),

$$\overline{g} = \frac{\delta n}{n} - \frac{\delta B}{B}, \qquad \overline{(v_\perp^2/v_{\rm th}^2)\,g} = -\frac{Z}{\tau}\frac{\delta n}{n} - 2\Big(1+\frac{1}{\beta_i}\Big)\frac{\delta B}{B}, \qquad \overline{(\cdots)} \equiv \frac{1}{n_i}\int \mathrm{d}^3v\,(\cdots). \tag{4}$$

These relations are algebraic: $\delta n/n$ and $\delta B/B$ at a point are linear functionals of g at that point, with coefficients that do not depend on the wavenumber. At finite $k_\perp\rho_i$ they contain gyroaveraging operators.

We expand g in Laguerre polynomials of $v_\perp^2/v_{\rm th}^2$ with coefficients $G_l(v_\parallel)$ (S09 eq. 205; CMM eq. B8; appendix A). The source term in (3) is linear in $v_\perp^2$, so the coefficients with $l \geq 2$ obey $\mathrm{d}G_l/\mathrm{d}t + v_\parallel\nabla_\parallel G_l = 0$ (CMM eq. B10; S09 eqs 210 and 212). The coefficients with $l = 0, 1$ are coupled through $\delta n/n$ and $\delta B/B$; the combinations $G^\pm$ of CMM's eqs (B5)–(B6) (S09 eqs 183–184) obey CMM's eq. (B4) (S09 eq. 181),

$$\frac{\mathrm{d}G^\pm}{\mathrm{d}t} + v_\parallel\nabla_\parallel G^\pm = \frac{v_\parallel F_M(v_\parallel)}{\Lambda^\pm}\,\nabla_\parallel\int G^\pm(v_\parallel')\,\mathrm{d}v_\parallel', \tag{5}$$

[2] S09's §§2.3 and 2.5 write the solutions with $\zeta^\mp = 0$ as $\zeta^\pm = f^\pm(x,y,z\mp v_A t)$, the opposite direction. We checked ours against S09's eq. (21), CMM's equations and GANDALF's right-hand side.

where $F_M(v_\parallel) = \pi^{-1/2} v_{\rm th}^{-1} e^{-v_\parallel^2/v_{\rm th}^2}$ and

$$\Lambda^\pm = -\frac{\tau}{Z} + \frac{1}{\beta_i} \pm \Delta, \qquad \Delta = \Big[\Big(1+\frac{\tau}{Z}\Big)^2 + \frac{1}{\beta_i^2}\Big]^{1/2} \tag{6}$$

(CMM eq. B7; S09 eqs 182 and 204).[3] We call the right-hand side of (5), of strength $1/\Lambda^\pm$, the Landau coupling; in linear theory it produces the collisionless damping of the compressive fluctuations (S09, §6.2.2). The coefficients with $l \geq 2$ obey (5) with $1/\Lambda = 0$, and we call them the sector $\Lambda = \infty$. The compressive sector therefore consists of independent kinetic equations for G^+, G^- and each G_l with $l \geq 2$, all advected by the same Alfvénic fields. Since $\Delta > 1 + \tau/Z$ and $\Delta > 1/\beta_i$, $\Lambda^+ - 1 > \beta_i^{-1}$ and $\Lambda^- < -\tau/Z$, so $\Lambda^+ > 1$ and $\Lambda^- < 0$ for all $\beta_i, \tau/Z > 0$ (S09, below eq. 200); at $\beta_i = \tau = Z = 1$, $\Lambda^\pm = \pm\sqrt{5}$.

2.3 One branch in the variable u

We now drop the branch label, set $u = v_\parallel/v_{\rm th}$, and write $F_M(u) = \pi^{-1/2} e^{-u^2}$ and $M_k[G] = \int u^k G \, {\rm d}u$. Since $v_\parallel = su$ when $v_A = 1$, equation (5) becomes

$$\partial_t G + \{\Phi, G\} + s\,\nabla_\parallel (LG) = 0, \tag{7}$$

with the branch operator

$$LG = uG - \frac{uF_M}{\Lambda} M_0[G], \tag{8}$$

which acts on the velocity variable only; for $\Lambda = \infty$ it is multiplication by u. Because the field relations (4) are algebraic, L contains no wavenumber.

For given Φ and Ψ, equation (7) is linear in G. Its nonlinear terms are the advection $\{\Phi, G\}$ and the part $s\{\Psi, LG\}$ of the streaming, which follows the perturbed field lines. The Alfvénic fields obey (2) whatever G does, so the compressive sector is passive. At finite $k_\perp \rho_i$ this fails; the induction equation, for example, contains $\nabla_\parallel(\delta n/n)$ (CMM, §3.1).

The density and field-strength fluctuations are linear combinations of $M_0[G^+]$ and $M_0[G^-]$ (A3), the parallel ion flow involves $M_1[G^\pm]$, and the pressure perturbations involve $M_0[G^\pm]$ and $M_2[G^\pm]$ (appendix A); the sector $\Lambda = \infty$ contributes to none of these moments.

2.4 Hermite representation

Following CMM (eqs B27 and B30), we expand $G = \sum_{m\geq 0} g_m H_m(u) F_M(u)/\sqrt{2^m m!}$, so that $g_m = \int H_m \, G \, {\rm d}u/\sqrt{2^m m!}$ and $G/F_M^{1/2} = \sum_m g_m \psi_m(u)$ with the orthonormal Hermite functions ψ_m. The g_m are CMM's $\tilde{G}_m$, and $M_0[G] = g_0$, $M_1[G] = g_1/\sqrt{2}$. In this basis (7) is the hierarchy

$$\partial_t g + \{\Phi, g\} + s\,\nabla_\parallel (Ag) = 0, \qquad g = (g_0, g_1, g_2, \dots)^{\mathsf T}, \tag{9}$$

where the ladder matrix A, the matrix of L, is tridiagonal with zero diagonal,

$$A_{m,m+1} = A_{m+1,m} = \sqrt{\frac{m+1}{2}} \quad \text{except} \quad A_{10} = \frac{1 - 1/\Lambda}{\sqrt{2}}. \tag{10}$$

The weight matrix $\mathsf{P} = {\rm diag}(p_m)$, with $p_0 = 1 - 1/\Lambda$ and $p_m = 1$ for $m \geq 1$, symmetrises it, $\mathsf{P}A = (\mathsf{P}A)^{\mathsf T}$, and $\sum_m p_m g_m^2$ is the free-energy product $[G, G]$ of section 3. A truncation at $m = M$

[3] CMM's eq. (B16) prints the square root in (6) as $[(1+\tau/Z) + \beta_i^{-2}]^{1/2}$, without the square on $1+\tau/Z$; with the square, as in S09's eq. (204), their decompositions (B15) and (B17) hold, and without it they do not (appendix A).

needs a closure; by default we use the zero closure $g_{M+1} = 0$, which keeps the first $M+1$ rows and columns of A. GANDALF (Kanekar, 2025) integrates (2) in Elsasser form together with (9) for one branch, with optional dissipation, hypercollisions and forcing, truncated with the zero closure or with the copy closure $g_{M+1} = g_{M-1}$ (section 4.4). We checked to round-off that its right-hand side without dissipation and hypercollisions is (9)–(10), for both signs of Λ and both closures.

The rest of the paper concerns one equation (7) with $\Lambda \notin [0, 1]$, which we call a branch whether it describes $G^{\pm}$ or a Laguerre coefficient with $l \geq 2$. Sections 3 to 5 treat a collisionless and unforced branch except where stated; sections 6 and 7 add forcing, dissipation and hypercollisions. The Hermite spectrum of the free energy is $W(m) = p_m \langle g_m^2 \rangle$, where $\langle \cdot \rangle$ is the average over the periodic box, so that $\sum_m W(m) = \langle [G, G] \rangle$.

3 Phase-velocity slices

This section shows that each branch (7) is a continuum of passive scalars, or slices, labelled by the parallel phase velocity. The Alfvénic fields Φ and Ψ are given functions of $\boldsymbol{r}$ and t; (2) is not used.

3.1 Without the Landau coupling

For $\Lambda = \infty$ the operator L is multiplication by u, and (7) holds separately at every u:

$$\partial_t G + \{\Phi, G\} + su\, \nabla_\parallel G = 0.$$

Since $\nabla_\parallel = \partial_z + \{\Psi, \cdot\}$ and $\{f, G\} = (\hat{\boldsymbol{z}} \times \nabla_\perp f) \cdot \nabla G$, this is the advection equation $\partial_t G + \boldsymbol{U}_u \cdot \nabla G = 0$ with the velocity

$$\boldsymbol{U}_u = \hat{\boldsymbol{z}} \times \nabla_\perp (\Phi + su\Psi) + su\, \hat{\boldsymbol{z}} = \boldsymbol{u}_E + su\, \hat{\boldsymbol{b}}, \qquad \nabla \cdot \boldsymbol{U}_u = 0. \tag{11}$$

At each u, G is a passive scalar, carried by the $\boldsymbol{E} \times \boldsymbol{B}$ flow and streaming along the perturbed field at the speed su, which is the motion of the guiding centre of a free ion with $v_\parallel = u v_{\mathrm{th}}$. Values of G at different u do not interact. This is the exact equation for the Laguerre coefficients G_l with $l \geq 2$ (S09, eqs 210 and 212).

3.2 With the Landau coupling

For finite Λ, $LG = uG - (uF_M/\Lambda)\, M_0[G]$ ties every velocity to the density $M_0[G]$, through the parallel electric field and the mirror force, so G at fixed u is no longer a passive scalar. The slices survive in other variables. The reason is that L acts on velocity only and is the same at every point and at every wavenumber, because the field relations (4) are algebraic. A change of velocity variables that diagonalises L therefore does so everywhere at once, and it passes through the advection and the streaming unchanged.

The argument is clearest for the hierarchy (9) truncated at $m = M$ with the zero closure, in which L is the constant $(M+1) \times (M+1)$ matrix A. Let $\boldsymbol{w}_j$ be a left eigenvector of A, $\boldsymbol{w}_j^{\mathsf{T}} A = u_j \boldsymbol{w}_j^{\mathsf{T}}$. Since $\boldsymbol{w}_j$ does not depend on position or time, it passes through ∂_t, ∂_z and both brackets, and multiplying (9) by $\boldsymbol{w}_j^{\mathsf{T}}$ gives for $a_j = \boldsymbol{w}_j^{\mathsf{T}} g$

$$\partial_t a_j + \{\Phi, a_j\} + su_j\, \nabla_\parallel a_j = \partial_t a_j + \{\Phi + su_j \Psi,\, a_j\} + su_j\, \partial_z a_j = 0.$$

Each a_j is a passive scalar, advected with the stream function $\Phi + su_j\Psi$ and streaming along z at the speed su_j, as G at fixed u was for $\Lambda = \infty$, and nothing couples a_j to the other a_l. The

nonlinearity enters only through the bracket with $\Phi + s u_j \Psi$, and the Landau coupling does not appear, because it has been absorbed into $\boldsymbol{w}_j$. For the a_j to be a change of variables, the u_j must be real and the $\boldsymbol{w}_j$ linearly independent, so that the a_j determine g. Both hold because $\mathsf{P}A$ is symmetric (section 2.4): A is self-adjoint in the free-energy product $\sum_m p_m g_m h_m$, which is positive definite when $p_0 = 1 - 1/\Lambda > 0$, that is, for $\Lambda \notin [0, 1]$. Section 4.4 normalises the a_j and treats both closures.

At $M = \infty$ the same steps need the eigenfunctions of L itself. Self-adjointness carries over: L is self-adjoint in the free-energy product

$$[f, g] = \int \frac{f g}{F_M} \, \mathrm{d}u - \frac{M_0[f]\, M_0[g]}{\Lambda}, \tag{12}$$

which is $\sum_m p_m f_m g_m$ in the Hermite variables and is positive definite for $\Lambda \notin [0, 1]$, as for every compressive branch of KRMHD ($\Lambda^+ > 1$, $\Lambda^- < 0$; section 2). Unlike A, however, L has no eigenfunctions of finite free energy: its spectrum is the whole real line and is continuous. Its generalised eigenfunctions are the Case–Van Kampen modes F_u of the branch (van Kampen, 1955; Case, 1959), with $L F_u = u F_u$ for every real u. Each is a beam of particles at $v_\parallel = u v_{\mathrm{th}}$, a δ-function in velocity, dressed by the response of the Maxwellian, through the Landau coupling, to the density of the beam (appendix B). The projection $\mathcal{T}[G](u) = [F_u, G]$ on F_u takes the place of $\boldsymbol{w}_j^{\mathsf{T}} g$, and its unit-weight form is the slice amplitude

$$a_u = \sqrt{\varrho(u)}\, [F_u, G], \qquad \varrho(u) = \frac{F_M(u)}{|\mathcal{K}(u + \mathrm{i}0)|^2}, \tag{13}$$

where $\mathcal{K}$ is the dielectric function of the branch, which contains the plasma dispersion function, and ϱ is the spectral density of the branch, positive with $\int \varrho \, \mathrm{d}u = 1/p_0$. For $\Lambda = \infty$, F_u is the bare beam, $\mathcal{K} = 1$, $\varrho = F_M$ and $a_u = G/\sqrt{F_M}$. The map $G \mapsto a_u$ is a fixed linear change of variables in velocity. It is invertible, divides the free energy among phase velocities without cross terms, and turns L into multiplication by u:

$$[G, G] = \int a_u^2 \, \mathrm{d}u, \qquad G \mapsto a_u \ \text{ takes } \ LG \mapsto u\, a_u \tag{14}$$

(theorem 4, which also gives a_u in terms of G and its Hilbert transform in u). Its inverse is, up to normalisation, the G-transform that Morrison and Pfirsch (1992) and Morrison (2000) introduced for linearised Vlasov–Poisson; Heninger and Morrison (2018) applied it to electrostatic drift kinetics, where the transformed components remain coupled through the potential.

3.3 The slice theorem

Theorem 1. *Assume that (i) $\Lambda \notin [0, 1]$, the value $\Lambda = \infty$ included; (ii) $\Phi(\boldsymbol{r}, t)$ and $\Psi(\boldsymbol{r}, t)$ are real functions on $\mathbb{T}^3 \times [0, t_1]$, with $\mathbb{T}^3$ a periodic box, whose gradients are continuous and Lipschitz in $\boldsymbol{r}$, uniformly in t; they need not satisfy (2) and may obey resistive or forced RMHD, or be prescribed; (iii) G and uG are continuously differentiable functions of $(\boldsymbol{r}, t)$ with values in the space $\mathcal{W}$ of functions of u with finite $\int G^2 F_M^{-1} \, \mathrm{d}u$ (appendix B). Then G solves (7) if and only if, at every $(\boldsymbol{r}, t)$ and for almost every u,*

$$\partial_t a_u + \boldsymbol{U}_u \cdot \nabla a_u = \partial_t a_u + \{\Phi + s u \Psi,\, a_u\} + s u\, \partial_z a_u = 0, \tag{15}$$

with $\boldsymbol{U}_u$ of (11).

Proof. This is the argument for the truncated hierarchy in section 3.2, with the map $\mathcal{A} : G \mapsto a$ of (13) in place of $\boldsymbol{w}_j^{\mathsf{T}}$. The map $\mathcal{A}$ is bounded and injective (theorem 4) and acts on u only, through a kernel independent of $\boldsymbol{r}$ and t, so it commutes with ∂_t, with spatial derivatives and with multiplication by functions of $(\boldsymbol{r}, t)$, hence with $\{\Phi, \cdot\}$, $\{\Psi, \cdot\}$ and $\nabla_\parallel$. With $\mathcal{A}LG = u\,\mathcal{A}G$ this gives $\mathcal{A}[\partial_t G + \{\Phi, G\} + s\,\nabla_\parallel(LG)] = \partial_t a + \{\Phi, a\} + su\,\nabla_\parallel a$, and $u\,\nabla_\parallel a_u = u\,\partial_z a_u + \{u\Psi, a_u\}$ gives (15). Since $\mathcal{A}$ is linear and injective, G solves (7) if and only if a solves (15). □

With the Landau coupling, therefore, each slice is again a passive scalar carried by $\boldsymbol{U}_u$, in three dimensions, and slices with different u do not interact. The parallel electric field and the mirror force, which enter through the Landau coupling, only dress the slices: the distribution at fixed u is not a passive scalar, but its projection a_u on F_u is. Since $\boldsymbol{U}_u$ is divergence-free, each slice keeps its own free energy $\int a_u^2\,\mathrm{d}^3\boldsymbol{r}$, and by (14) the nonlinearity moves no free energy between phase velocities (section 4.1). In terms of the Elsasser potentials, the stream function of the slice at u is

$$\Phi + su\Psi = \tfrac{1}{2}\left[(1+su)\,z^+ + (1-su)\,z^-\right], \tag{16}$$

so the slices at $su = -1$ and $+1$ are advected by z^- alone and by z^+ alone, and the slice at $u = 0$, the distribution at $v_\parallel = 0$, by Φ alone. The sectors have different transforms, but $\boldsymbol{U}_u$ depends only on u, so at each u the slices of G^+, G^- and every G_l with $l \geq 2$ are carried by the same flow:

$$(\partial_t + \boldsymbol{U}_u \cdot \nabla)(a_u^+,\, a_u^-,\, a_u^{(2)},\, a_u^{(3)}, \dots) = 0. \tag{17}$$

The compressive sector of KRMHD is therefore a continuum of passive scalars labelled by phase velocity.

3.4 Dissipation, collisions and forcing

Let the right-hand side of (7) be $-\mathcal{B}G - \mathcal{C}G + F$, where the linear operators $\mathcal{B}$ and $\mathcal{C}$ act on $\boldsymbol{r}$ alone and on u alone, respectively, and F is a source. The proof of theorem 1 gives

$$\partial_t a_u + \boldsymbol{U}_u \cdot \nabla a_u = -\mathcal{B}\,a_u - \int \hat{\mathcal{C}}(u, u')\,a_{u'}\,\mathrm{d}u' + \sqrt{\varrho(u)}\;\mathcal{T}[F](u), \tag{18}$$

where $\hat{\mathcal{C}}$ is the kernel of $\mathcal{A}\,\mathcal{C}\,\mathcal{A}^{-1}$. Damping at a rate that depends only on $k_\perp$ or k_z, applied with the same coefficient to every Hermite moment, is of the form $\mathcal{B}$ and damps each slice separately. With $e_m = H_m(u)F_M(u)/\sqrt{2^m m!}$ and $r_m(u)$ the mth Hermite coefficient of F_u, a polynomial of degree m with $r_0 = 1$ and $r_1 = \sqrt{2}\,u$, one has $\mathcal{T}[e_m](u) = p_m\,r_m(u)$, and the functions $\phi_m = \sqrt{p_m \varrho}\;r_m$ form an orthonormal basis of $L^2(\mathbb{R}, \mathrm{d}u)$. For $\mathcal{C}e_m = \nu_m e_m$, as for hypercollisions $\nu_m = \nu(m/M)^n$, the kernel is $\hat{\mathcal{C}}(u, u') = \sum_m \nu_m\,\phi_m(u)\,\phi_m(u')$, which couples different slices unless all ν_m are equal; collisions are the only term in (18) that exchanges free energy between slices. Forcing of g_0 by a field $f(\boldsymbol{r}, t)$, $F = f\,e_0$, gives every slice the same spatial pattern, with amplitude $(1 - 1/\Lambda)\sqrt{\varrho(u)}\,f$.

3.5 Slices in a turbulent run

Figure 1 releases a density blob, $g(0) = f\,e_0$, into decaying Alfvénic turbulence with $\chi = 1.00$ and follows the 17 slices of the hierarchy truncated at $M = 16$. A density perturbation loads every slice with the same pattern f (section 3.4). Each slice is then carried by its own stream function $\Phi + su_j\Psi$, which at $su_j = -1.0218$, 0 and $+1.0218$ is mainly z^-, Φ alone and mainly z^+, and streams along z at su_j. The three slices end visibly different, and g_0 mixes them. The slices keep their values as they are carried: on the plane $z = 0$ the peaks of the slices at $su_j = \pm 1.0218$ are

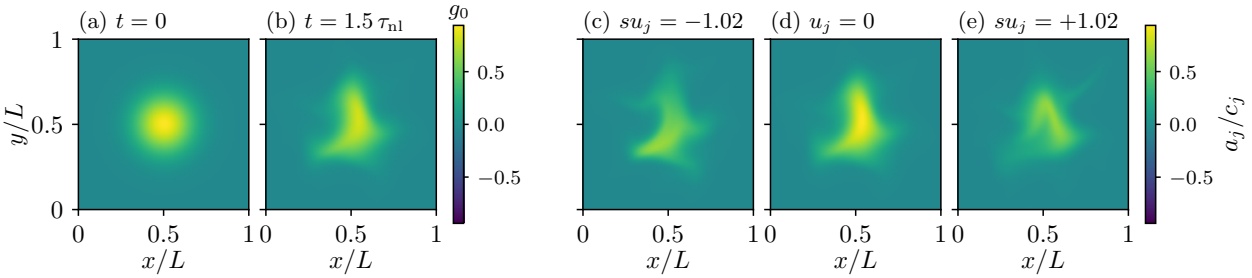


Figure 1: A density blob in decaying Alfvénic turbulence, on the plane $z = 0$ (coordinates in units of the box side $L = 1$). (a, b) The density moment g_0 at $t = 0$ and $t = 1.5\,\tau_{\rm nl}$. (c–e) Three of the 17 slices a_j of the run (section 3.2) at $t = 1.5\,\tau_{\rm nl}$, with $su_j = -1.0218$, $u_j = 0$ and $su_j = +1.0218$; their stream functions $\Phi + su_j\Psi$ are mainly z^-, Φ alone and mainly z^+. At $t = 0$ each slice is a constant multiple $c_j f$ of the blob f, and the panels show a_j/c_j, so all three started as panel (a). They differ at $t = 1.5\,\tau_{\rm nl}$ because each was carried by its own flow, and g_0 in (b) is a weighted sum of all 17 slices. Panels (c) and (e) are also fainter than (d), because those slices stream along z at su_j: the material on their plane $z = 0$ started on the planes $z = +0.21$ and $z = -0.21$, where f is about 0.7 of its value at $z = 0$. Each group of panels has one symmetric colour scale. GANDALF run on a 64^3 grid with $M = 16$ (zero closure), $\Lambda = \sqrt{5}$ and $\beta_i = 1$; balanced Gaussian random Elsasser fields with $1 \le k_\perp/k_1 \le 3$, $|k_z|/k_1 \le 2$ and $\chi = 1.00$; hyper-resistivity damps $z^\pm$ and every g_m; no forcing or hypercollisions; f is a Gaussian of standard deviation $1/8$ in x and y and $1/4$ in z; $\tau_{\rm nl} = 0.1357$ is the outer eddy time.

0.69 of the initial peak, close to the factor 0.71 by which f falls over their streaming distance of 0.21 along z, and the peak of the slice at $u_j = 0$ is 0.998 of it. GANDALF's $\partial_t g$ projected on the slices agrees with (26), computed with brackets that do not use GANDALF, to 5.2×10^{-16} of the largest slice rate, while advecting every slice by Φ alone gives relative errors of 0.30 to 1.46. Over the run each slice energy changes only by the hyper-resistive loss of that slice, to 1.2×10^{-6} of its initial energy, so no free energy passes between slices.

4 Free energy and the Hermite spectrum in slice variables

This section collects the consequences of theorem 1 for the free energy and its Hermite spectrum. Without collisions, dissipation or sources the energy of every slice is conserved, so the free energy is not moved between phase velocities, and free energy moves between Hermite moments only as the correlations between slices change.

4.1 Slice energies

Let $\mathcal{E}(u) = \langle a_u^2 \rangle$ be the slice energy, so that $\frac{1}{2} n_i T_i\, \mathcal{E}(u)\, \mathrm{d}u$ is the free energy per unit volume in the slices with phase velocities in $[u, u + \mathrm{d}u]$ and $\int \mathcal{E}\, \mathrm{d}u = \langle [G, G] \rangle = \sum_m W(m)$ by (14). Under the hypotheses of theorem 1, with no collisions, dissipation or sources acting on G, (11) and $\nabla \cdot \boldsymbol{U}_u = 0$ give $\partial_t a_u^2 + \nabla \cdot (\boldsymbol{U}_u\, a_u^2) = 0$, so $\mathcal{E}(u)$ is independent of time for every u. Every functional $\int \varphi(u)\, \mathcal{E}(u)\, \mathrm{d}u$ is then conserved. The free energy $W^\pm$ of each branch (S09) and the two further quadratic invariants found by Chandran et al. (2026), $\Gamma^\pm$ and the phase-space helicity $\mathcal{H}^\pm$ (their eqs B25 and B18), are the members $\varphi = 1$, u and $1/u$, up to normalisation; we do not use the last two here.

The nonlinearity of KRMHD therefore moves no free energy between phase velocities. Of the terms that act on G in a simulation, damping at a rate that depends only on $k_\perp$ or k_z, applied alike to every Hermite moment, removes free energy from each slice separately (18), and only velocity-dependent collisions exchange it between slices.

4.2 The two-slice law and the Hermite flux

Let $C(u,u') = \langle a_u\, a_{u'}\rangle$ be the two-slice correlation, whose diagonal is $\mathcal{E}(u)$.

Proposition 1 (Two-slice law)**.** *Under the hypotheses of theorem 1, with no collision operator, source or dissipation acting on G, and for slice amplitudes that are continuously differentiable in $\boldsymbol{r}$ and t with derivatives continuous in u,*

$$\frac{\mathrm{d}C(u,u')}{\mathrm{d}t} = -s\,(u-u')\,\langle a_{u'}\,\nabla_\parallel a_u\rangle. \tag{19}$$

Proof. By (15), $\partial_t a_u = -\{\Phi, a_u\} - su\,\nabla_\parallel a_u$, and the Leibniz rule gives $\partial_t(a_u a_{u'}) = -\{\Phi, a_u a_{u'}\} - su'\,\nabla_\parallel(a_u a_{u'}) - s\,(u-u')\,a_{u'}\nabla_\parallel a_u$. Since $\nabla\cdot\boldsymbol{u}_E = \nabla\cdot\hat{\boldsymbol{b}} = 0$, the first two terms are divergences, whose box averages vanish. □

The $\boldsymbol{E}\times\boldsymbol{B}$ flow carries the two slices of a pair together and cannot change their correlation, which changes only through their relative streaming along the perturbed field. Phase mixing and unmixing are therefore the decorrelation and recorrelation of slices along $\nabla_\parallel$, a definition that refers to no wavenumber.

Free energy moves between Hermite moments only through the streaming term. In the Hermite form (9) the bracket does not change $\langle g_m^2\rangle$, and since $\mathsf{P}A$ is symmetric and $\langle g_n\nabla_\parallel g_{n'}\rangle$ antisymmetric, only the coupling between m and $m+1$ survives in $E_{\le m} = \sum_{n\le m} W(n)$:

$$\frac{\mathrm{d}E_{\le m}}{\mathrm{d}t} = -\Pi_{m+1/2}, \qquad \Pi_{m+1/2} = s\,b_m\langle g_m\nabla_\parallel g_{m+1} - g_{m+1}\nabla_\parallel g_m\rangle, \qquad b_m = p_m A_{m,m+1}, \tag{20}$$

the free energy per unit time and volume that passes from moment m to $m+1$. In slice variables $\Pi_{m+1/2} = -\iint K_m(u,u')\,\partial_t C(u,u')\,\mathrm{d}u\,\mathrm{d}u'$, with the Christoffel–Darboux kernel $K_m(u,u') = \sum_{n\le m}\phi_n(u)\,\phi_n(u')$ of the functions ϕ_n of section 4.3, which by (19) is a weighted integral of the pair correlations $\langle a_{u'}\nabla_\parallel a_u\rangle$. The flux must be computed with the full $\nabla_\parallel$: the Hermite flux of Meyrand et al. (2019) uses it, whereas a flux computed with ∂_z alone, which we call the k_z-only flux, omits the part of the streaming that follows the tilt of the field lines and does not close the budget below.

With forcing of g_0 at the rate ε, dissipation that acts on every g_m in the same way and hypercollisions, the free energy in each moment obeys

$$\frac{\mathrm{d}W(m)}{\mathrm{d}t} = \Pi_{m-1/2} - \Pi_{m+1/2} + \varepsilon\,\delta_{m0} - D_\perp(m) - D_\nu(m), \qquad \Pi_{-1/2} = 0, \tag{21}$$

where $D_\perp(m)$ and $D_\nu(m)$ are the rates at which the spatial dissipation and the hypercollisions remove free energy from moment m. The brackets and the hyper-resistivity of GANDALF act on all moments alike, so (21) holds for its equations up to the error of the dealiased brackets, with the flux (20) evaluated with the same brackets; in the simulations of section 7 it closes to better than $10^{-3}\,\varepsilon$.

4.3 The Hermite spectrum and the two-slice correlation

With the orthonormal basis $\phi_m = \sqrt{p_m \varrho}\, r_m$ of $L^2(\mathbb{R}, \mathrm{d}u)$ (section 3.4), $g_m = p_m^{-1/2} \int \phi_m\, a_u\, \mathrm{d}u$, and the Hermite spectrum is a quadratic form in the two-slice correlation,

$$W(m) = \iint C(u,u')\, \phi_m(u)\, \phi_m(u')\, \mathrm{d}u\, \mathrm{d}u'. \tag{22}$$

If the slices are completely decorrelated, $C(u,u') = e(u)\, \delta(u-u')$, then $W(m) = \int e\, \phi_m^2\, \mathrm{d}u$. At $\Lambda = \infty$, ϕ_m is the Hermite function ψ_m, and for $e = F_M$

$$\int F_M\, \psi_m^2\, \mathrm{d}u = \frac{\Gamma(m+\frac{1}{2})}{\sqrt{2}\,\pi\, m!} = \frac{1}{\pi\sqrt{2m}}\Big(1 - \frac{1}{8m} + O(m^{-2})\Big), \tag{23}$$

where $\Gamma(\cdot)$ is Euler's gamma function. This is the $m^{-1/2}$ spectrum of linear phase mixing. For one Fourier mode whose density moment is driven by white noise that injects free energy at the rate ϵ_k, the exact steady state of Kanekar et al. (2015, eq. 4.10) is the spectrum of decorrelated slices with $e = \pi(1-1/\Lambda)^2 \epsilon_k\, \varrho/(s|k_z|)$; at $\Lambda = \infty$

$$W(m) = \frac{\pi\epsilon_k}{s|k_z|}\, \frac{\Gamma(m+\frac{1}{2})}{\sqrt{2}\,\pi\, m!} \;\simeq\; \frac{\varepsilon}{s|k_z|\sqrt{2m}} \quad (m \gg 1), \tag{24}$$

and at every Λ the large-m form is $\varepsilon/(s|k_z|\sqrt{2m})$, with ε the free-energy injection rate, and the flux (20) is ε through every m. Density forcing loads every slice with the same spatial pattern (section 3.4), two slices loaded together dephase at the rate $s|k_z(u-u')|$, and the steady state superposes all past injections, so the slices are decorrelated in the mean. Free energy injected at $m = 0$ thus moves up the ladder at the speed $\mathrm{d}m/\mathrm{d}t = \Pi/W = s|k_z|\sqrt{2m}$ and reaches $m \simeq (sk_z t)^2/2$ after a time t.

Correlated slices give steeper spectra. If $C(u,u') = [e(u)e(u')]^{1/2}\, c(u-u')$ at $\Lambda = \infty$, with e slowly varying, the large-m form of ψ_m gives

$$W(m) \simeq \frac{\hat{c}(\sqrt{2m})}{\pi\sqrt{2m}} \int e\, \mathrm{d}u, \qquad \hat{c}(q) = \int c(\delta)\cos(q\delta)\, \mathrm{d}\delta, \tag{25}$$

so $\sqrt{2m}$ acts as a wavenumber in velocity space (Schekochihin et al., 2016; Adkins and Schekochihin, 2018). A power law $\hat{c} \propto q^{-p}$ gives $W \propto m^{-(p+1)/2}$, and a Lorentzian correlation $c(\delta) = \delta_c/[\pi(\delta^2+\delta_c^2)]$ of width δ_c gives $\hat{c} = e^{-q\delta_c}$ and $W(m) \propto m^{-1/2}\exp(-\sqrt{2m}\,\delta_c)$. A spectrum that falls exponentially in $\sqrt{m}$ is therefore one in which slices further apart than δ_c in phase velocity are decorrelated and nearer ones are not.

4.4 Truncated hierarchies

Codes such as GANDALF evolve the moments $g_0, \ldots, g_M$ of a branch and close the hierarchy at $m = M$. For the zero closure A is the leading $(M+1)\times(M+1)$ block of the ladder matrix (10); the copy closure $g_{M+1} = g_{M-1}$ changes, for $M \geq 2$, only $A_{M,M-1}$, to $\sqrt{M/2} + \sqrt{(M+1)/2}$. Section 3.2 gave the argument for part (ii) of the following theorem; the theorem adds the normalisation of the slices, the copy closure and the relation to the moments.

Theorem 2. *Let $\Lambda \notin [0,1]$, the value $\Lambda = \infty$ included, and use the zero closure or, for $M \geq 2$, the copy closure.*

(i) *A has $M+1$ real and simple eigenvalues $u_0 < \cdots < u_M$. The right eigenvector of u_j is $\boldsymbol{r}_j = (r_0(u_j), \ldots, r_M(u_j))^{\mathsf{T}}$, with r_m the polynomials of section 3.4; the left eigenvector is $\boldsymbol{w}_j = \mathsf{P}\boldsymbol{r}_j$, and $\boldsymbol{w}_j^{\mathsf{T}}\boldsymbol{r}_l = n_j\,\delta_{jl}$ with $n_j > 0$.*

(ii) *For arbitrary real Φ and Ψ, and brackets evaluated by any bilinear rule, possibly followed by linear operations, such as dealiasing, that act on every g_m alike, g solves (9) if and only if the slice amplitudes $a_j = \boldsymbol{w}_j^{\mathsf{T}} g/\sqrt{n_j} = \sum_m p_m R_{mj} g_m$, with $R_{mj} = r_m(u_j)/\sqrt{n_j}$, satisfy*

$$\partial_t a_j + \{\Phi + s u_j \Psi,\, a_j\} + s u_j\, \partial_z a_j = 0, \qquad j = 0, \ldots, M. \tag{26}$$

(iii) *$g_m = \sum_j R_{mj}\, a_j$, and $\sum_m p_m |g_m(\boldsymbol{k})|^2 = \sum_j |a_j(\boldsymbol{k})|^2$ for every Fourier mode.*

Proposition 2. *For the zero closure, the u_j are the zeros of r_{M+1}, and with $\lambda_j = 1/n_j$*

$$\sum_{j=0}^{M} \lambda_j\, f(u_j) = \int \varrho(u)\, f(u)\, \mathrm{d}u \qquad \text{for every polynomial } f \text{ of degree at most } 2M+1: \tag{27}$$

the u_j and λ_j are the nodes and weights of the Gauss rule of the spectral density $\varrho\,\mathrm{d}u$, and $a_j = \sqrt{\lambda_j}\,\mathcal{T}[G](u_j)$ for states in the span of the retained moments.

Appendix C gives the proofs. A truncated hierarchy is therefore exactly $M+1$ passive slices at the Gauss nodes of the Case–Van Kampen measure, and the results above hold for it with sums over j in place of integrals over u. At $\Lambda = \infty$ the nodes are the Gauss–Hermite nodes (Parker and Dellar, 2015; Tassi, 2022); in KRMHD the Landau coupling enters only A_{10} of a matrix that does not depend on the wavenumber, and the Alfvénic fields do not depend on g, so the decoupling is exact in three dimensions. The largest phase velocity $u_{\max} = u_M$ hardly depends on Λ; with $q_m = (2m+1)^{1/2}$,

$$\begin{aligned} u_{\max} &= q + 2^{-1/3}\alpha_1\, q^{-1/3} - \frac{2^{-2/3}\alpha_1^2}{10}\, q^{-5/3} + O(q^{-3}), && q = q_{M+1} \quad \text{(zero closure)},\\ u_{\max} &= q + 2^{-1/3}\alpha_1'\, q^{-1/3} - 2^{-5/3}\Big(\frac{4}{5\alpha_1'} + \frac{\alpha_1'^2}{5}\Big) q^{-5/3} + O(q^{-3}), && q = q_M \quad \text{(copy closure)}, \end{aligned} \tag{28}$$

where $\alpha_1 = -2.3381074$ and $\alpha_1' = -1.0187930$ are the first zeros of the Airy function and of its derivative (Szegő, 1975; Olver, 1959, theorem 6.32); $u_{\max} = 7.25$ at $M = 32$. Slice j is advected with the stream function $\Phi + s u_j \Psi$, so an explicit time step for the nonlinear term must resolve the speed $\max|\nabla_\perp \Phi| + s u_{\max} \max|\nabla_\perp \Psi|$, not $\max|\nabla_\perp \Phi| + s \max|\nabla_\perp \Psi|$; the simulations of section 7 set their time step in this way (appendix D.1).

5 Frozen field lines

When the field lines are frozen into the $\boldsymbol{E} \times \boldsymbol{B}$ flow, which is common to all slices (11), each slice streams, in the frame of the flow, along field lines that do not move. This Lagrangian linearity is due to S09 (§6.3), who showed with the Cauchy solution for the perturbed field that in the frame of $\boldsymbol{u}_E$ the derivative $\hat{\boldsymbol{b}} \cdot \nabla$ is taken along the field lines at $t = 0$ (their eqs 214–216), and concluded that the compressive fluctuations do not cascade along the moving field lines; Kunz et al. (2015, §§2.6 and 4.1) restated it for pressure-anisotropic plasmas, and Lithwick and Goldreich (2001, §2.3.1) made the corresponding remark for a passive scalar in MHD turbulence. We give the solution operator and the hypotheses, and use volume preservation to obtain exact statements about volume-integrated statistics such as $W(m)$.

5.1 The frozen-line theorem

We treat one branch at a time, with state $g(\boldsymbol{r},t)$: the function $G(\boldsymbol{r},\cdot,t)$ of u, or $(g_0,\dots,g_M)$ in a truncated hierarchy, where A replaces L. With collisions $-\mathcal{C}g$ as in (18), (7) reads

$$\partial_t g + \{\Phi, g\} + s\,\nabla_\parallel(Lg) = -\mathcal{C}g. \tag{29}$$

Let $D_t = \partial_t + \{\Phi,\cdot\}$ and $\nabla_\parallel(t) = \partial_z + \{\Psi(\cdot,t),\cdot\}$, let $\boldsymbol{X}_t$ be the flow map of $\boldsymbol{u}_E$, with $\partial_t \boldsymbol{X}_t(\boldsymbol{\xi}) = \boldsymbol{u}_E(\boldsymbol{X}_t(\boldsymbol{\xi}),t)$ and $\boldsymbol{X}_0(\boldsymbol{\xi}) = \boldsymbol{\xi}$, and let $\mathcal{X}_t h = h \circ \boldsymbol{X}_t^{-1}$ turn a function of the labels $\boldsymbol{\xi}$ into a function of position. The first of equations (2) states that the residual $R = \partial_t\Psi + \{\Phi,\Psi\} - \partial_z\Phi$ vanishes. The hypotheses are

(A1) *Ideal induction.* On $[0,t_1]$ the residual R depends on z and t only. The vorticity equation may be forced, viscous or prescribed.

(A2) *Kinetic equation.* The branch has $\Lambda \notin [0,1]$, and g obeys (29), where $\mathcal{C}$ is a linear operator on the velocity variable, the same at every point and independent of the fields, and is bounded or dissipative. There is no spatial dissipation of g and no source.

(A3) *Smoothness.* Φ, Ψ and all their spatial derivatives are continuous on $\mathbb{T}^3\times[0,t_1]$, except that Φ may jump at finitely many times.

(30)

Under (A3), $\boldsymbol{X}_t$ is a volume-preserving diffeomorphism of $\mathbb{T}^3$. The frozen generator $L_0 = s\,\nabla_\parallel(0)\,L + \mathcal{C}$ describes the linear problem with the field lines fixed in their initial shape and no flow, and e^{-tL_0} is its solution operator.

Theorem 3 (Frozen-line theorem)**.** *Assume (30), and let g be a classical solution of (29) on $[0,t_1]$. Then*

$$g(t) = \mathcal{X}_t\, e^{-tL_0}\, g(0), \qquad 0 \le t \le t_1. \tag{31}$$

Proof. With $\hat{\boldsymbol{b}} = \hat{\boldsymbol{z}} + \hat{\boldsymbol{z}}\times\nabla_\perp\Psi$, $\partial_t\hat{\boldsymbol{b}} - \nabla\times(\boldsymbol{u}_E\times\hat{\boldsymbol{b}}) = \hat{\boldsymbol{z}}\times\nabla R$, which vanishes under (A1), so $\hat{\boldsymbol{b}}$ obeys the ideal induction equation and, since $\nabla\cdot\boldsymbol{u}_E = \nabla\cdot\hat{\boldsymbol{b}} = 0$, has Cauchy's solution (Webb et al., 2014, example 3) $\hat{\boldsymbol{b}}(\boldsymbol{X}_t(\boldsymbol{\xi}),t) = (\partial\boldsymbol{X}_t/\partial\boldsymbol{\xi})\,\hat{\boldsymbol{b}}(\boldsymbol{\xi},0)$. The chain rule then gives, for a differentiable function h of the labels,

$$\nabla_\parallel(t)\,\mathcal{X}_t h = \mathcal{X}_t\,\nabla_\parallel(0) h; \tag{32}$$

equivalently, by the Jacobi identity,

$$[D_t, \nabla_\parallel]\, f = \{R, f\}, \tag{33}$$

which vanishes under (A1). Write $g(t) = \mathcal{X}_t\hat{g}(t)$. By (29), $\partial_t\hat{g} = -[s\nabla_\parallel(t)(Lg) + \mathcal{C}g]\circ\boldsymbol{X}_t$. The operators L and $\mathcal{C}$ act at each point with kernels independent of $\boldsymbol{r}$ (for L because the field relations of KRMHD are algebraic), so they commute with composition by $\boldsymbol{X}_t$, and (32) gives $\partial_t\hat{g} = -L_0\hat{g}$ with $\hat{g}(0) = g(0)$. For $\Lambda \notin [0,1]$, $s\nabla_\parallel(0)L$ is antisymmetric in the free-energy norm, which gives uniqueness. □

Without collisions, (31) decouples into slices,

$$a_u(t) = \mathcal{X}_t\,\mathsf{S}_{sut}\,a_u(0), \qquad \mathsf{S}_\ell = e^{-\ell\nabla_\parallel(0)}, \tag{34}$$

where S_ℓ moves a function a distance ℓ along the field lines of $\hat{\boldsymbol{b}}(0)$: each slice is carried rigidly along the initial field lines at the speed su and then relabelled by the flow. This makes exact, for the slices, the remark of S09 (§6.2.4) that the ballistic response is also the nonlinear solution in Lagrangian variables. Whether solutions of the ideal three-dimensional RMHD equations stay smooth is not known in general, so (A3) is an assumption on $[0,t_1]$.

5.2 Single-point statistics and forcing

Let O be a function of the local state, all the velocity-space information at one point, and of finitely many of its derivatives along the field. We call

$$\mathcal{O}_t[f] = \int O(f, \nabla_\parallel(t) f, \nabla_\parallel(t)^2 f, \dots)\, \mathrm{d}^3 r \tag{35}$$

a single-point statistic. Volume preservation and $\nabla_\parallel(t)^k \mathcal{X}_t = \mathcal{X}_t \nabla_\parallel(0)^k$ give $\mathcal{O}_t[\mathcal{X}_t y] = \mathcal{O}_0[y]$.

Corollary 1 (Single-point statistics)**.** *Assume (30), and let $h(t) = e^{-tL_0} g(0)$. Then $\mathcal{O}_t[g(t)] = \mathcal{O}_0[h(t)]$ for every single-point statistic (35) and $0 \le t \le t_1$: every volume-integrated single-point statistic of g takes the value it has in the linear problem along the field lines frozen at $t = 0$, with no flow.*

The corollary covers $W(m,t)$, the slice correlations $C(u,u';t)$, the collisional dissipation rate and the Hermite flux (20) with the full $\nabla_\parallel$. It does not cover spectra in $k_\perp$ or k_z, fluxes resolved in $k_\perp$ shells or computed with ∂_z, which involve Fourier filters that do not commute with $\mathcal{X}_t$. With a source F, Duhamel's formula in the labels shows that the source injected at t' phase-mixes along the field lines frozen at t' and is then relabelled by the flow. For forcing that is white in time this gives the following.

Corollary 2 (White forcing)**.** *Let the forcing be $F\,\mathrm{d}t = \mathrm{d}Y$, with Y a Wiener process with spatially smooth realisations and covariance operator Σ whose increments are independent of the past of g and of the Alfvénic fields, let (A1) and (A3) hold for every realisation, and let g otherwise satisfy (A2). For every single-point statistic whose integrand is a quadratic form,*

$$\mathrm{E}\,\mathcal{O}_t[g(t)] = \mathrm{E}\,\mathcal{O}_0\big[e^{-tL_0} g(0)\big] + \int_0^t \mathrm{E}\,\mathcal{O}_{t'}\big[e^{-(t-t')L_{t'}} \xi\big]\, \mathrm{d}t', \tag{36}$$

where $L_{t'} = s\,\nabla_\parallel(t')\,L + \mathcal{C}$ and ξ is a random field with zero mean and covariance Σ, independent of all other variables. The mean of $W(m,t)$ is therefore the sum of the linear responses to the initial condition and to the individual injections, each phase-mixing along the field lines frozen at its injection time.

Sketch of proof. In the labels, $\mathrm{d}\hat{g} = -L_0 \hat{g}\,\mathrm{d}t + \mathcal{X}_t^{-1}\mathrm{d}Y$, whose solution is an Itô integral with an integrand fixed by the past. The cross term between the initial condition and the stochastic integral has zero mean, the Itô isometry turns the rest into $\int_0^t \mathrm{E}\,\mathcal{O}_0[e^{-(t-t')L_0}\mathcal{X}_{t'}^{-1}\chi]\,\mathrm{d}t'$, and relabelling with (32) gives (36). □

The corollary concerns the mean: a single realisation contains cross terms between injections at different times, and for forcing correlated in time the mean cross terms between injections closer than the correlation time survive.

5.3 Field-aligned wavenumbers

Since $\hat{\boldsymbol{b}}(t)$ is smooth and divergence-free, $K_\parallel(t) = -\mathrm{i}\,\nabla_\parallel(t)$ is self-adjoint on $L^2(\mathbb{T}^3)$ (e.g. Kato, 1966). Its spectral variable is the field-aligned wavenumber, and in a uniform field its spectral projections are the Fourier filters in k_z.

Corollary 3 (Field-aligned wavenumbers)**.** *Assume (30), and let $E_t(I)$ be the spectral projection of $K_\parallel(t)$ on a Borel set I. Then*

$$E_t(I)\, g(t) = \mathcal{X}_t\, e^{-tL_0}\, E_0(I)\, g(0). \tag{37}$$

For real g the parts with positive and negative field-aligned wavenumber evolve separately, so no part of g changes the sign of its field-aligned wavenumber, and without collisions the field-aligned spectrum of every slice is constant in time.

Proof. By (32) and the unitarity of $\mathcal{X}_t$, $E_t(I) = \mathcal{X}_t E_0(I) \mathcal{X}_t^{-1}$, and $E_0(I)$ commutes with e^{-tL_0}. Without collisions e^{-tL_0} multiplies the component of the slice at u at field-aligned wavenumber $k_\parallel$ by $e^{-\mathrm{i}suk_\parallel t}$. □

The Eulerian k_z of the same part of g can change sign. In the frozen-in shear $\Phi = -Szy$, $\Psi = -Sty$, an exact but not periodic solution of (2) with $\nabla_\parallel(t) = \partial_z + St\,\partial_x$, a Fourier mode of the labels becomes a Fourier mode of position with

$$k_z(t) = k_z - Stk_x, \qquad k_\parallel(t) = k_z(t) + Stk_x = k_z. \tag{38}$$

Its k_z changes sign at $t = k_z/(Sk_x)$ while its field-aligned wavenumber stays fixed, so a diagnostic that sorts the free energy by the sign of k_z counts the mode as reversed, although nothing has changed along the field.

5.4 What breaks the theorem

Resistivity or hyper-resistivity on Ψ, parallel dissipation of $z^\pm$, and forcing of Ψ in general make R depend on x and y, so that (33) does not vanish and the field lines are not carried by the flow. Dissipation of g that depends on $k_\perp$ or k_z does not commute with the relabelling and violates (A2). Fourier truncation does not commute with it either; its error vanishes as the resolution increases at fixed t, but not uniformly in t. In the equations that GANDALF integrates, the hyper-resistivity acts alike on $z^\pm$ and every g_m and violates (A1) and (A2), the hypercollisions satisfy (A2), and forcing that adds the same increment to z^+ and z^- keeps (A1).

5.5 A numerical check

The runs of figure 2 integrate one ideal initial state at $N = 16$ to 64 for about three eddy times and compare each run with the frozen-line prediction $e^{-tL_0} g(0)$, computed with GANDALF's right-hand side for g with $\Phi = 0$ and $\Psi = \Psi(0)$ held fixed, and with the k_z-only prediction ($\Phi = \Psi = 0$). For the statistics covered by corollary 1 the relative difference falls by two to three orders of magnitude from $N = 16$ to 64, to between 1.5×10^{-5} for $W(m)$ and 1.4×10^{-4} for the field-aligned flux, while for $W(k_\perp)$, $W(|k_z|)$ and the flux with ∂_z it stays between 0.13 and 0.38 at every N. The k_z-only prediction misses $W(m)$ by 15% and the field-aligned flux by 37%. In ideal turbulence forced through Φ, the response of $W(m)$ to each white injection into g converges with N to phase mixing along the field lines frozen at its injection time, as corollary 2 states. With one term added, the departure from the frozen-line prediction still falls with N for viscosity or forcing of Φ, but not for resistivity or forcing of Ψ or for damping of g in $k_\perp$ or k_z.

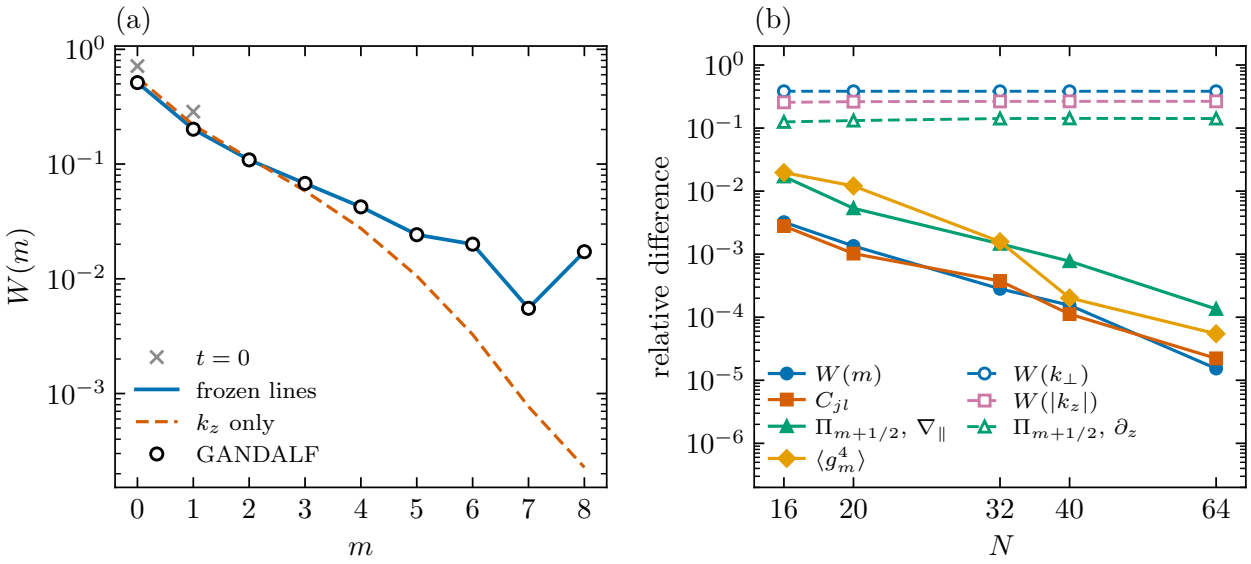


Figure 2: Test of the frozen-line theorem. Ideal GANDALF runs (no dissipation, forcing or collisions) from one initial state at N^3 grid points, with $M = 8$ (zero closure), $\Lambda = \sqrt{5}$, $\beta_i = 1$, random g_0 and g_1, $\chi = 2.21$, the term $\{\Phi, g_0\}$ 2.91 times the streaming term in rms, and $\Delta t = 0.0005$. (a) The Hermite spectrum $W(m)$ at $t = 0.125$ for $N = 64$: initial state (crosses), GANDALF (circles), frozen-line prediction (solid line) and k_z-only prediction (dashed line). (b) The relative difference between GANDALF and the frozen-line prediction at $t = 0.125$ against N, for the single-point statistics $W(m)$, the slice correlations $C_{jl} = \langle a_j a_l \rangle$ at the nodes of section 4.4, $\Pi_{m+1/2}$ with $\nabla_\parallel$ and $\langle g_m^4 \rangle$ (filled markers), and for $W(k_\perp)$, $W(|k_z|)$ and $\Pi_{m+1/2}$ with ∂_z (open markers).

5.6 Echoes

Under (30), $W(m, t)$ and the field-aligned Hermite flux evolve as in the linear problem $\partial_t h = -L_0 h$, so a return of free energy to low m must already be present there, as a linear echo of initial unmixing correlations, the recurrence of a truncated hierarchy, or slow phase mixing at small field-aligned wavenumbers; with white forcing the same holds for $\mathrm{E}\, W(m, t)$. In this sense ideal, smooth dynamics without spatial dissipation of g produces no stochastic echoes in $W(m)$. In drift kinetics with a straight guide field (Schekochihin et al., 2016; Parker et al., 2016) the particles stream along a fixed direction, $[D_t, \partial_z] f = -\{\partial_z \Phi, f\} \neq 0$, and the z-dependent flow couples positive and negative $k_\parallel$, which is the mechanism of the stochastic echo; a flow independent of z gives none (Schekochihin et al., 2016, appendix B; Kanekar, 2015, §5.2). In KRMHD the shear $\partial_z \boldsymbol{u}_E$ that would change $k_\parallel$ also bends the field lines together with g, so under (A1) the commutator (33) vanishes and no part of g moves between positive and negative field-aligned wavenumbers (corollary 3), although its Eulerian k_z can change sign (38).

The theorem does not contradict the simulations in which echoes were reported (Meyrand et al., 2019), which, like ours, dissipate both the magnetic field and the compressive distribution in space and so violate (A1) and (A2). An echo in $W(m)$ requires such terms, or a failure of deterministic flux freezing for rough fields as the resistivity goes to zero, which (A3) excludes; Eyink et al. (2013) reported such a failure at high conductivity, and Eyink (2018, §VIII.C) argued that in a turbulent inertial range the ideal Ohm's law does not imply flux freezing. The question for turbulence is therefore what the terms that break (30) do to $W(m)$, which the model of section 6 addresses.

6 A two-channel model of the Hermite spectrum

In a turbulent steady state the compressive distribution is forced and dissipated, and the hypotheses (30) fail in two ways: the induction equation is not ideal, and g is dissipated in space. The exact results still constrain what the nonlinearity can do. It moves no free energy between slices (section 4.1), and as long as the field lines stay frozen into the flow it moves none between field-aligned wavenumbers (corollary 3). This section builds a model on the assumption that the two failures matter only through the removal of free energy at small perpendicular scales, and lists the predictions by which it can be tested.

6.1 The two channels

Phase mixing at a frozen field-aligned wavenumber. With white forcing, the mean of $W(m)$ is a sum of linear responses to the individual injections, each phase mixing along the field lines frozen at its injection time (corollary 2). Each injection has, along those lines, a distribution of field-aligned wavenumbers κ, which it keeps (corollary 3). By (24), free energy injected at κ moves up the Hermite ladder at the speed $\mathrm{d}m/\mathrm{d}t = s\kappa\sqrt{2m}$ and carries the flux

$$\Pi_{m+1/2} \simeq s\kappa\sqrt{2m}\, W(m) \qquad (m \gg 1), \tag{39}$$

with no part returning to lower m.

A uniform sink. Each slice is a passive scalar, which the turbulence carries to the dissipation scale, where the spatial dissipation removes it, slice by slice (section 4.1). For passive scalars in turbulence the rate at which variance is dissipated stays finite as the diffusivity goes to zero (Shraiman and Siggia, 2000; Falkovich et al., 2001), and the time a scalar takes to reach the dissipation scale is set by the flow, not by the diffusivity. All slices share the $\boldsymbol{E}\times\boldsymbol{B}$ flow, and in its frame they differ only by their streaming along the field lines (theorem 3), which moves them along the lines but does not by itself refine them across the lines. We therefore assume that the turbulence cascades every slice in the same way. If the perpendicular spectrum of g_m then has the same shape for every m, the dissipation, which acts alike on every moment, removes free energy from each moment in proportion to its free energy,

$$D_\perp(m) = \frac{W(m)}{T}, \tag{40}$$

with one time T for all m. Equivalently, for single-point statistics such as $W(m)$, the nonlinearity of KRMHD acts as a damping at a rate $1/T$ that does not depend on m. Section 6.4 replaces it by a rate that depends on the age of the free energy, which the simulations require (section 7.5).

In the steady state of (21), treated as continuous in m and without collisions, the flux (39) changes with m only through the sink (40):

$$\frac{\mathrm{d}\Pi}{\mathrm{d}m} = -\frac{W(m)}{T} = -\frac{\Pi(m)}{s\kappa T\sqrt{2m}}. \tag{41}$$

Since $\mathrm{d}\sqrt{2m}/\mathrm{d}m = 1/\sqrt{2m}$, this integrates to $\ln[\Pi(m)/\Pi(0)] = -\sqrt{2m}/(s\kappa T)$. All the free energy enters the ladder through g_0, so $\Pi(0) = \varepsilon$, the rate of injection into g_0, and $W = \Pi/(s\kappa\sqrt{2m})$ gives

$$W(m) = \frac{\varepsilon}{s\kappa\sqrt{2m}} \exp\Big(-\frac{\sqrt{2m}}{s\kappa T}\Big). \tag{42}$$

The prefactor is the $m^{-1/2}$ spectrum of linear phase mixing, whose flux ε is the same at every m. The exponential can be read off by following the free energy up the ladder. By (39) it moves at $\mathrm{d}m/\mathrm{d}t = s\kappa\sqrt{2m}$, so $\mathrm{d}\sqrt{2m}/\mathrm{d}t = s\kappa$, and it reaches m at the age

$$\vartheta_m = \frac{\sqrt{2m}}{s\kappa}. \tag{43}$$

The sink removes it at the rate $1/T$ on the way, so the fraction that reaches m is $\exp(-\vartheta_m/T)$, which is the exponential in (42). Because the injected pattern is fixed in the Eulerian frame and the field lines are tilted, the injected free energy has a distribution $P(\kappa)$ of field-aligned wavenumbers, and

$$W(m) = \varepsilon \int P(\kappa)\, \frac{1}{s\kappa\sqrt{2m}} \exp\Big(- \frac{\sqrt{2m}}{s\kappa T} \Big)\, \mathrm{d}\kappa, \tag{44}$$

with each κ breaking at its own m, so that a broad $P(\kappa)$ turns the stretched exponential into a more gradual decline. For a single κ the local slope is $\mathrm{d}\ln W/\mathrm{d}\ln m = -\frac{1}{2}(1 + \sqrt{m/m_*})$, which passes through -1 at $m = m_*$ and $-3/2$ at $m = 4m_*$, so the spectrum is close to m^{-1} around the break without being a power law. A spread of κ makes the decline more gradual, because the free energy at large κ phase mixes fastest and survives to the largest m; for a distribution with a Gaussian tail of width σ, (44) falls at very large m as $\exp[-c\,(m/s^2T^2\sigma^2)^{1/3}]$ with c of order one, more slowly than the single-κ exponential. With the distribution of the injected pattern measured at $\beta_i = 1$ and $M = 64$ (section 7; mean $1.12\,k_1$, with k_1 the wavenumber of the box) and m_* evaluated at the mean, the local slope of (44) stays within 0.15 of -1 over a factor of five in m around m_* and reaches only -2 at $16\,m_*$ and -2.4 at $64\,m_*$, against -2.5 and -4.5 for a single κ. A spectrum measured over a limited range near m_* can therefore look like a power law somewhat steeper than m^{-1}. Hypercollisions add the sink $D_\nu(m)$ to (41). For the comparison with simulations we solve, for each κ, the forced linear hierarchy (9) with $\nabla_\parallel$ replaced by $\mathrm{i}\kappa$, with the damping rate $1/(2T)$ added to every amplitude and with the hypercollisions of the run, and sum over the measured $P(\kappa)$ (appendix D). This reproduces (44) at large m and is exact for the linear problem at small m. Once T and $P(\kappa)$ are measured, the model has no free parameter.

The form (42) is that of Zocco and Schekochihin (2011, eq. 112) for the Hermite spectrum of growing modes, with their 2γ replaced by $1/T$, and Schekochihin et al. (2016, eqs 2.58–2.59) solved the same continuity equation in m with the collisional sink $2\nu m$ in place of $1/T$. What the model adds is the identification of the sink with the passive-scalar cascade, which the exact results of sections 4 and 5 make the only effect of the nonlinearity on $W(m)$ once the field lines are frozen.

6.2 Scales

The exponential in (42) matters once the age (43) of the free energy reaches the cascade time T. The spectrum therefore follows $m^{-1/2}$ up to the break

$$m_* = \frac{(s\kappa T)^2}{2}, \tag{45}$$

and beyond it decays exponentially in $\sqrt{m}$, not as a power law.

The dependence of m_* on the plasma parameters follows from estimating the two times at the outer scale. The field-aligned wavenumber κ is S09's $k_{\parallel 0}$ (§6.3), the wavenumber along the perturbed field, which without a parallel cascade stays that of the large-scale stirring. It is therefore of the order of the parallel wavenumber $k_\parallel$ of the outer-scale turbulence. Phase mixing then carries free energy to moment m in the time

$$\vartheta_m \sim \frac{\sqrt{m}}{k_\parallel v_{\mathrm{th}}}, \tag{46}$$

and the perpendicular cascade removes it in a time of the order of the outer eddy-turnover time,

$$T \sim \frac{1}{k_\perp u_\perp}, \tag{47}$$

where $u_\perp$ is the $\boldsymbol{E} \times \boldsymbol{B}$ velocity at the outer scale. The free energy reaches m before the cascade removes it, $\vartheta_m \lesssim T$, as long as

$$\sqrt{m} \lesssim \frac{k_\parallel v_{\mathrm{th}}}{k_\perp u_\perp} = \frac{v_{\mathrm{th}}}{v_A} \frac{k_\parallel v_A}{k_\perp u_\perp} = \frac{\beta_i^{1/2}}{\chi}, \tag{48}$$

where $v_{\mathrm{th}}/v_A = \beta_i^{1/2}$ and $\chi = k_\perp u_\perp/(k_\parallel v_A)$ is the nonlinearity parameter. Hence

$$m_* \sim \frac{\beta_i}{\chi^2}. \tag{49}$$

This is the phase-mixing threshold $k_\parallel v_{\mathrm{th}} \sim \sqrt{m}\, k_\perp u_\perp$ of Schekochihin et al. (2016, eq. 4.5), written in KRMHD variables, with its coefficient fixed by (45) and the cascade time; for $\chi \sim 1$ it is the crossover $m \sim \beta_i$ stated by Kawazura et al. (2019). The estimate takes T to be a fixed multiple of the outer eddy time $1/(k_\perp u_\perp)$; section 7.4 finds that m_* grows somewhat faster than χ^{-2} down to $\chi \approx 0.56$ and about as χ^{-4} between 0.56 and 0.40, because the sink per outer eddy time is not fixed (section 7.5). If the collisional cutoff m_c lies below m_*, most of the injected free energy reaches it and is dissipated by collisions; if it lies above, most is dissipated at small perpendicular scales, and the collisional share is about $\exp(-\sqrt{2m_c}/(s\kappa T))$.

In slice variables the break is a correlation width. Two slices loaded with the same pattern dephase at the rate $s\kappa|u - u'|$ while the sink removes them at the rate $1/T$, so slices closer than $1/(s\kappa T) = 1/\sqrt{2m_*}$ in phase velocity stay correlated, and by (25) a correlation of this width gives the exponential in (42).

6.3 Predictions

The model predicts the following, each measurable independently of the others.

(i) *Uniform sink.* $D_\perp(m)/W(m)$ is independent of m, and the budget (21) closes with the flux (20) computed with the full $\nabla_\parallel$, but not with the k_z-only flux.

(ii) *Shape.* With T from (i) and $P(\kappa)$ measured along field lines for the injected pattern, $W(m)$ and the field-aligned flux follow the model. A return flux would make both fall below it at large m, and would make the flux fall below $s\sqrt{2(m+1)}\,\kappa_1(m)\,W(m)$, the flux that $W(m)$ carries at its measured field-aligned wavenumber. A sink that grows with the age of the free energy (section 6.4) also lowers W and the flux at large m, but not their ratio to this flux.

(iii) *Dependence on β_i.* At fixed χ the Alfvénic turbulence does not depend on β_i, so $P(\kappa)$ is unchanged and, if T is too, $m_* \propto \beta_i$: the break moves into a fixed range of m as β_i decreases. With a sink that depends on the age of the free energy (section 6.4), the effective T falls somewhat as β_i decreases, because the same moments then hold older free energy.

(iv) *Anomaly.* T stays finite and does not depend on the dissipation coefficients as the resolution increases. Section 7.6 tests this at four resolutions.

(v) *Frozen wavenumber.* The field-aligned spectrum of the free energy stays at the wavenumbers of the injected pattern and does not change as the inertial range lengthens with the resolution.

The echo picture (Meyrand et al., 2019) predicts instead a return flux along the field, which reduces the net field-aligned flux and $W(m)$ below the model at large m, and a field-aligned spectrum of g that extends to larger wavenumbers as the inertial range lengthens, as g inherits the parallel cascade of the Alfvénic fields.

6.4 Limits of the model

The model rests on two assumptions that the exact results do not guarantee. First, it assumes that the field lines stay frozen long enough for the injected κ to be kept, although the hyper-resistivity of any simulation, and resistivity in a plasma, reconnect them. Meyrand et al. (2019) argued that this relabelling gives g the parallel structure of the Alfvénic fields within about one eddy-turnover time. Second, it assumes that the perpendicular cascade has a dissipative anomaly. Drivas and Eyink (2017) showed that anomalous dissipation of a passive scalar requires Lagrangian spontaneous stochasticity, the non-uniqueness of trajectories as the diffusivity goes to zero, and Eyink (2018, §VIII.C) argued that the same stochasticity breaks flux freezing in a turbulent inertial range. If the two go together, the frozen κ of the first assumption must fail at the scales where the anomaly of the second operates. The model can survive this only if the free energy that phase mixes sits at scales where the field lines are still effectively frozen and the relabelling acts on free energy that the perpendicular sink is already removing. Section 7.3 measures the field-aligned spectrum of the free energy at four resolutions to see whether this holds. The model also takes the cascade rate to be the same for every slice; slices with large $|u|$ stream fast along wandering field lines, which may refine them across the lines faster than slow slices, and prediction (i) measures how large this effect is. In our runs at $\beta_i = 0.3$ the field-aligned spectrum of the free energy is the same from 64^3 to 256^3 grid points, apart from a tail holding less than 0.5% of it (section 7.3), while the sink rate falls by 18–35% with each doubling of the resolution (section 7.6). At the resolutions reached here the first assumption therefore holds and the second is not confirmed.

If the sink depends on the time ϑ since injection rather than being uniform, as section 7.5 finds, free energy injected at κ reaches m at the age $\vartheta = \sqrt{2m}/(s\kappa)$, and (42) becomes

$$W(m) = \frac{\varepsilon}{s\kappa\sqrt{2m}} \exp\Big(- \int_0^{\sqrt{2m}/(s\kappa)} \gamma(\vartheta)\, \mathrm{d}\vartheta \Big), \tag{50}$$

with $\gamma(\vartheta)$ the rate at which the perpendicular cascade removes free energy of age ϑ. For a passive scalar injected at the outer scale, γ is small at first, while the cascade carries the fresh pattern towards the dissipation scale, and approaches a constant once the scalar's spectrum has filled in. A uniform $1/T$ then removes too much free energy at small m and too little at large m. The function $\gamma(\vartheta)$ belongs to the Alfvénic turbulence, not to β_i, so at fixed χ one $\gamma(\vartheta)$ should describe every β_i.

7 Simulations

We test the predictions of section 6.3 in forced simulations. The runs differ in β_i, in the nonlinearity parameter χ, and in the numbers of Hermite moments M and of grid points N^3. Table 1 summarises what they find, and the subsections give the evidence.

7.1 Set-up

We integrate (2) in Elsasser form with one compressive branch (9), the branch Λ^+ of (6) with $\tau = Z = 1$, truncated at $m = M$ with the zero closure, using GANDALF (Kanekar, 2025) in

Table 1: The predictions of section 6.3 and what the simulations find.

Prediction	Result	Section
(i) Uniform sink	Close to uniform at $\beta_i = 1$, rising by 20% over $1 \le m \le 16$; at 64^3 and $\chi \approx 1$ the sink is one function of the age of the free energy at every β_i	7.2, 7.5
(ii) Shape	$W(m)$ within 8% for $m \le 30$ and the field-aligned flux within 3–7% for $m \le 24$ at $\beta_i = 1$; at $\beta_i = 0.1$, $W(m)$ lies up to 42% above the model with a uniform sink; no return flux in any run	7.2, 7.5
(iii) Dependence on β_i	$m_* \propto \beta_i^{1.09\pm0.04}$ for $0.05 \le \beta_i \le 0.3$ at $\chi \approx 0.95$; against χ faster than χ^{-2}, because the sink depends on the age of the free energy and, at $\chi = 0.40$, weakens per outer eddy time	7.4, 7.5
(iv) Anomaly	Not confirmed: at $\beta_i = 0.3$, T grows by 5–6 outer eddy times with each doubling of the resolution up to 256^3	7.6
(v) Frozen wavenumber	At $\beta_i = 0.3$ the field-aligned spectrum of the free energy is the same from 64^3 to 256^3, apart from a tail holding less than 0.5% of it, while that of the Alfvénic velocity extends with the resolution	7.3

double precision. The box is periodic with side $L = 1$, so $k_1 = 2\pi$; $v_A = 1$ and $\tau_A = L/v_A = 1$. Appendix D.1 gives the time stepping, the averaging windows and the checks of every run.

The Alfvénic fluctuations are forced through Φ alone, so that z^+ and z^- receive the same increment, which keeps the forcing out of the induction equation (section 5). The forcing is white in time and acts on the Fourier modes with $1 \le k_\perp/k_1 \le 2$ and $|k_z| = k_1$, with injection rate $\varepsilon_A = 0.3$ except in the runs at lower χ (section 7.4). Excluding $k_z = 0$ from the band avoids the two-dimensional condensate that a band including it drove at this resolution. The density moment g_0 is forced on the same band, white in time, with its amplitude rescaled at every step so that the expected injection rate is $\varepsilon = 0.3$; the band excludes $k_\perp = 0$, where injected free energy would phase mix with no perpendicular cascade. Every injection therefore has the single Eulerian parallel wavenumber $k_z = \pm k_1$, and its field-aligned wavenumbers differ from k_1 only through the tilt of the field lines.

GANDALF applies the hyper-resistivity $\eta\,(k_\perp^2/k_{\perp,\max}^2)^2$ alike to $z^\pm$ and to every g_m, with $\eta = 300$ except at lower χ; there is no parallel dissipation. Hypercollisions damp g_m at the rate $\nu\,(m/M)^6$ for $m \ge 2$ and so remove its free energy at the rate $2\nu\,(m/M)^6$. The attenuation they impose on free energy streaming up the ladder depends on $\nu\sqrt{M}/s$, so we set $\nu = 30\,\beta_i^{1/2}(32/M)^{1/2}$, which keeps it fixed as β_i and M change. Free energy moving up the ladder at $\mathrm{d}m/\mathrm{d}t = s\kappa\sqrt{2m}$ passes the moments $m \le M/2$ quickly enough that the hypercollisions remove only 3% of the injected free energy there at $\beta_i = 1$ and $M = 32$; above $m = M/2$ they remove most of what arrives.

We characterise the outer scale by the part of Φ in the shells $k_\perp/k_1 \le 2.5$: its energy-weighted mean $k_\perp$ and $|k_z|$, $k_{\perp,\mathrm{out}}$ and $k_{\parallel,\mathrm{out}}$, and its rms $\boldsymbol{E}\times\boldsymbol{B}$ velocity u_{out} give the nonlinearity parameter $\chi = k_{\perp,\mathrm{out}}u_{\mathrm{out}}/k_{\parallel,\mathrm{out}}$ and the outer eddy time $\tau_{\mathrm{nl}} = 1/(k_{\perp,\mathrm{out}}u_{\mathrm{out}})$. In the run at $\beta_i = 1$, 64^3 and $M = 32$, $k_{\perp,\mathrm{out}} = 1.55\,k_1$, $k_{\parallel,\mathrm{out}} = 1.11\,k_1$ and $u_{\mathrm{out}} = 0.71$, so $\chi = 0.99$ (0.85 with the rms field-aligned wavenumber of the same field in place of $|k_z|$) and $\tau_{\mathrm{nl}} = 0.144$. The Alfvénic sector does not depend on β_i, and its outer scale hardly depends on the resolution (section 7.6), so every run of table 2 has χ between 0.92 and 0.99. At 64^3 the inertial range is short: in the run at $\beta_i = 1$,

Table 2: The forced runs at $\chi \approx 1$, on N^3 grid points with M Hermite moments. The window is the averaging interval in units of τ_A; $1/T$ is the median of $D_\perp(m)/W(m)$ over $2 \le m \le 16$, the same range in every run, below the collisional range $m > M/2$. The checks of every run are in table 5.

β_i	N	M	window	$1/T$	$T/\tau_{\rm nl}$
1	64	32	3–53	0.437	15.9
0.3	64	32	15.5–53	0.478	13.5
0.1	64	32	16–66	0.590	11.2
1	64	64	16–66	0.421	15.4
0.3	64	64	16–66	0.508	13.2
0.1	64	64	16–66	0.604	11.0
0.3	96	64	16–66	0.403	16.4
0.3	128	64	16–66	0.352	18.4
0.3	256	64	10–60	0.257	24.6
1	128	64	16–66	0.284	22.6
1[a]	64	64	16–66	0.427	15.3

[a] g_1 forced in place of g_0, on $|k_z| = k_1$ and $2k_1$ (section 7.2).

$E_A(k_\perp)$ follows $k_\perp^{-5/3}$ only over $2 \le k_\perp/k_1 \le 8$ and is steeper than $k_\perp^{-4}$ from $k_\perp/k_1 = 9$. Table 2 lists the runs at $\chi \approx 1$.

The energy change of every step is split into the parts due to the ideal dynamics, the forcing, the hyper-resistivity and the hypercollisions. This gives $D_\perp(m)$ and $D_\nu(m)$ in (21) exactly for every moment. Every 20 steps we record $W(m)$, its perpendicular spectrum, the field-aligned flux (20) and the k_z-only flux, the split of the k_z-only flux into phase-mixing and unmixing parts by the sign of k_z (Schekochihin et al., 2016), and $E_A = \frac{1}{2}\langle|\nabla_\perp\Phi|^2 + |\nabla_\perp\Psi|^2\rangle$ and its spectrum. Field-aligned wavenumbers are measured along traced field lines, as is done for the Alfvénic fields (Cho and Vishniac, 2000; Meyrand et al., 2019). On snapshots spaced by $5\,\tau_A$ within the averaging window ($10\,\tau_A$ at 96^3 and above, and 10–$50\,\tau_A$ in the runs of the m_* scan) we trace 4096 lines of $\hat{\boldsymbol{b}} = \hat{\boldsymbol{z}} + \hat{\boldsymbol{z}} \times \nabla_\perp\Psi$ over four box lengths in z, sample g_m, Φ and Ψ along them, and compute the spectra along the lines; κ_1 denotes the first moment of $|\kappa|$. The distribution $P(\kappa)$ of the injected free energy is measured in the same way, on fresh realisations of the density forcing sampled along the same lines (appendix D).

7.2 No return flux

In every run the free energy phase mixes at its injected field-aligned wavenumber, with no return flux. We show this first for the run at $\beta_i = 1$, 64^3 and $M = 32$. Its Hermite spectrum is marginally resolved, with 1.1% of the free energy at $m > 3M/4$, and its break lies far beyond $m = M$ (section 7.4), so we treat it as a pilot that tests predictions (i) and (ii) but not the break.

The perpendicular dissipation removes free energy from each moment nearly in proportion to its free energy, as (40) assumes. $D_\perp(m)/W(m)$ lies between 0.37 and 0.47 for $0 \le m \le 16$, with a systematic rise of 20% over $1 \le m \le 16$ that section 7.5 traces to the age of the free energy (figure 3b). We take $1/T = 0.437$, its median over $2 \le m \le 16$, so $T = 2.29 = 15.9\,\tau_{\rm nl}$. The budget (21) closes to $4 \times 10^{-4}\,\varepsilon$ in every moment with the field-aligned flux, and only to 8% with the k_z-only flux, as prediction (i) states.

With this T and the $P(\kappa)$ of the injected pattern, the model has no free parameter. It reproduces

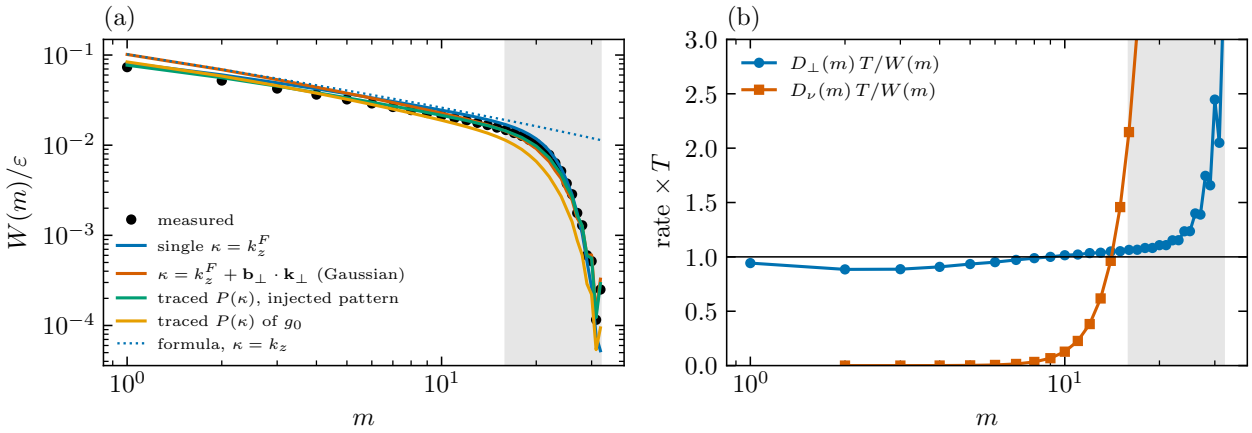


Figure 3: (a) The Hermite spectrum $W(m)/\varepsilon$ at $\beta_i = 1$ (black dots) and the two-channel model with the measured T and four choices of κ: a single $\kappa = k_z^F = k_1$, the wavenumber of the forcing (blue); $\kappa = k_z^F + \boldsymbol{b}_\perp \cdot \boldsymbol{k}_\perp$ with a Gaussian distribution of the tilt (red); the distribution $P(\kappa)$ of the injected pattern traced along field lines (green); and the distribution of g_0 itself traced along field lines (yellow). The dotted line is the formula (42) with $\kappa = k_z$ and no collisions. (b) The rates at which the perpendicular dissipation (circles) and the hypercollisions (squares) remove free energy from moment m, divided by $W(m)$ and multiplied by T. The shading marks $m > M/2$, where the hypercollisions act.

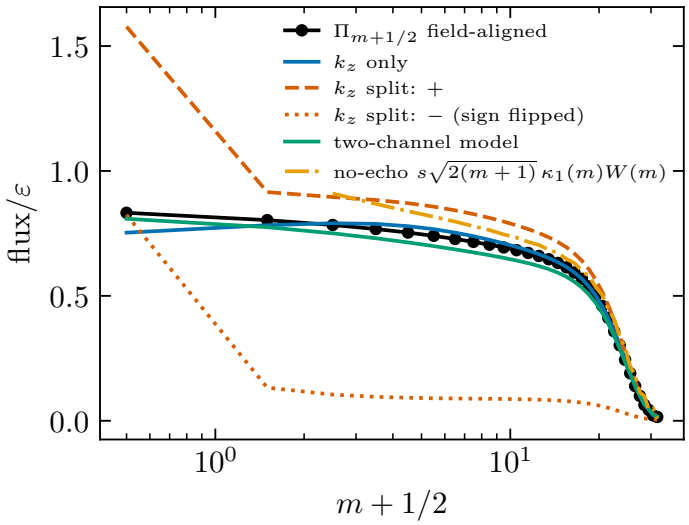


Figure 4: Hermite fluxes divided by ε at $\beta_i = 1$: the field-aligned flux (20) (black), the k_z-only flux (blue), its phase-mixing part (dashed) and its unmixing part with the sign reversed (dotted), the two-channel model (green), and the flux $s\sqrt{2(m+1)}\,\kappa_1(m)\,W(m)$ that $W(m)$ would carry if all of it phase mixed at the measured field-aligned wavenumber (dash-dotted).

$W(m)$ to within 8% for $1 \le m \le 30$ (figure 3a) and the field-aligned flux to within 3–7% for $m \le 24$ (figure 4); for $1 \le m \le 16$ the measured flux lies 4–7% above the model. The moments g_m have κ_1 between $1.07\,k_1$ and $1.19\,k_1$ for $1 \le m \le 21$, with no trend in m, against $1.14\,k_1$ for the injected pattern, and the flux corresponds to phase mixing at $\Pi_{m+1/2}/[s\sqrt{2(m+1)}\,W(m)] = 1.02$–$1.10\,k_1$ for $3 \le m \le 23$, so there is no room for a return flux of more than a few per cent. Only in the last moments, $m \ge 27$, where the hypercollisions dominate, does the flux fall below the model, by up to 12%, and W by 14% at $m = 32$; a sink that grows with the age of the free energy acts in this direction. The tilt of the field lines matters: with a single $\kappa = k_1$ the model gives W 15–16% too high for $2 \le m \le 16$, and with a Gaussian distribution of the tilt in place of the traced one the measured $W(m)$ ranges from 0.72 to 1.22 times the model. With the field-aligned distribution of g_0 itself, whose mean is $0.61\,k_1$ because free energy at small κ leaves $m = 0$ slowly, it gives W too low by a factor of up to 1.9 at $m = 24$: the free energy phase mixes at the wavenumbers at which it was injected.

A split of the k_z-only flux into phase-mixing and unmixing parts by the sign of k_z (Schekochihin et al., 2016) reports a backward flux where the field-aligned flux has none: at $\beta_i = 1$ its unmixing part is -0.11 to -0.15 times its phase-mixing part for $1 \le m \le 16$ (figure 4), and at $\beta_i = 0.3$ it grows from about -0.13 times the phase-mixing part at 64^3 to -0.27 to -0.35 at 256^3, while the field-aligned flux shows no deficit at any resolution. The split assigns to the backward direction the forward flux of modes whose k_z and field-aligned wavenumber have opposite signs (38), and such modes become more common as the tilt of the field lines grows with the resolution. In every run of table 2 the per-m budget closes to better than $5 \times 10^{-4}\,\varepsilon$ with the field-aligned flux, and only to 2–8% with the k_z-only flux.

In the other runs of table 2 neither $W(m)$ nor the field-aligned flux falls increasingly below the model at large m (sections 7.5 and 7.6). In the runs of the m_* scan (section 7.4), whose collision-free range reaches much older free energy, the flux falls relative to the model with a uniform sink as m grows, to as much as 37% below it at $m = 256$ at $\chi = 0.40$, as a sink that grows with the age of the free energy makes it. The flux there is still the one that $W(m)$ carries at its measured field-aligned wavenumber: it lies between 0.85 and 1.00 times $s\sqrt{2(m+1)}\,\kappa_1(m)\,W(m)$ for $8 \le m \le M/2$ in every run of the scan, without the deficit growing with m that a return flux would cause. No run shows a return flux.

Meyrand et al. (2019) forced the compressive fluctuations through g_1, the parallel flow, on modes with $|k_z| \le 2k_1$. To see what this changes, we repeated the run at $\beta_i = 1$, 64^3 and $M = 64$ with g_1 forced in place of g_0, on the same band of $k_\perp$ but on $|k_z| = k_1$ and $2k_1$, at the same rate. The Alfvénic turbulence is the same, with $\chi = 0.95$ and $\tau_{\rm nl} = 0.153$, and so is the perpendicular sink, $1/T = 0.427$ against 0.421. The injected pattern has $\kappa_1 = 1.63\,k_1$ along the field lines, against $1.12\,k_1$. The spectrum is steeper at small m, with a local slope of -0.66 over $2 \le m \le 8$ against -0.54, and the same at larger m, -0.71 over $8 \le m \le 32$ against -0.72; it is nowhere close to m^{-1}. The model with the traced $P(\kappa)$, injected at $m = 1$, reproduces $W(m)$ to within 3.5% for $0 \le m \le 32$, and the field-aligned flux lies 2–4% above it for $m \ge 1$, with no deficit at large m, so forcing g_1 on two parallel wavenumbers adds no return flux. At $m = 16$ the field-aligned flux carries 75% of the injected free energy, against 66% with g_0 forced, and the hypercollisions remove 61% of it, against 50%, because the larger injected κ carries the free energy to the collisional range sooner; the model gives 60% and 48%.

7.3 The field-aligned wavenumber

The free energy keeps the field-aligned wavenumbers at which it was injected, at every resolution. The injected pattern has $k_z = k_1$ but $\kappa_1 = 1.14\,k_1$ along the field lines at $\beta_i = 1$ and $M = 32$

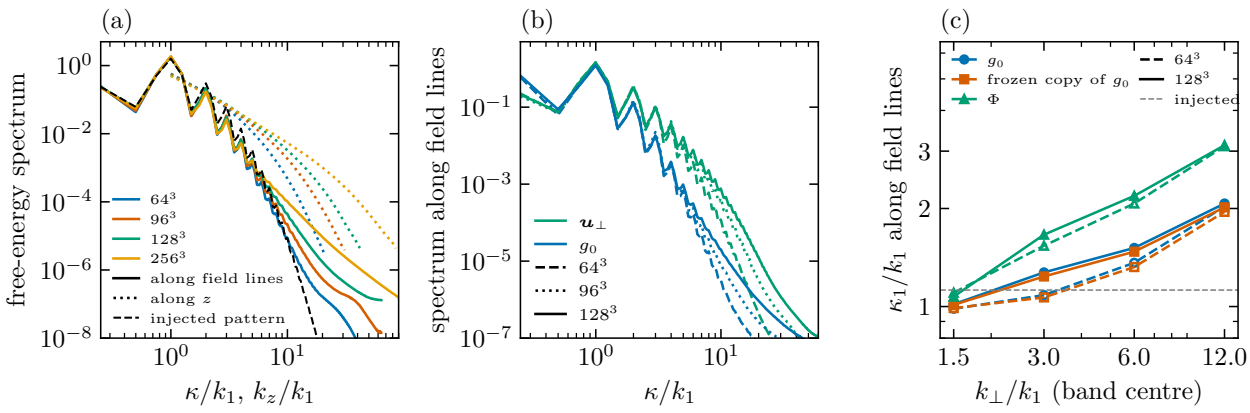


Figure 5: (a) The spectrum of the free energy, summed over m, along the traced field lines (solid) and along z (dotted), each normalised to unit integral, at $\beta_i = 0.3$ and $M = 64$ on 64^3, 96^3, 128^3 and 256^3 grid points, with the spectrum of the injected pattern along the same lines (dashed). (b) The spectra along the same lines of the whole $\boldsymbol{E} \times \boldsymbol{B}$ velocity $\boldsymbol{u}_\perp$ (green) and of g_0 (blue), each normalised to unit integral, at $\beta_i = 0.3$ on 64^3 (dashed), 96^3 (dotted) and 128^3 (solid) grid points. (c) The mean field-aligned wavenumber κ_1 of g_0 (circles), of its frozen copy (squares; appendix D.4) and of Φ (triangles) filtered into the bands $k_\perp/k_1 \in [0, 2]$, $[2, 4]$, $[4, 8]$ and $[8, 16]$, at $\beta_i = 0.3$ on 64^3 (dashed, open) and 128^3 (solid, filled) grid points; the dashed grey line is κ_1 of the injected pattern.

(1.10–1.14 k_1 in the runs at $\beta_i = 0.3$ below), because the lines are tilted: the rms of the tilt $\boldsymbol{b}_\perp \cdot \boldsymbol{k}_\perp$ over the pattern is $0.89\,k_1$, and the rms field-aligned wavenumber is $1.34\,k_1$.

Figure 5(a) shows the spectrum of the free energy, summed over m, along the field lines and along z in the runs at $\beta_i = 0.3$ and $M = 64$ on 64^3 to 256^3 grid points. Along the field lines it does not change with the resolution. At every resolution 3.7–4.4% of the free energy lies beyond $2\,k_1$, against 6.9–8.2% for the injected pattern, and its mean field-aligned wavenumber is 1.00–1.03 k_1; the free energy is weighted towards smaller κ than the injection, because free energy at larger κ reaches the collisional range sooner. Measured along z, the same free energy shows a spectrum that extends further with each doubling of the resolution, as a parallel cascade would: the share beyond $8\,k_1$ grows from 0.2% at 64^3 to 0.9%, 1.8% and 4.3% at 96^3, 128^3 and 256^3. A cascade along the field would extend the field-aligned spectrum in the same way. What grows along the field lines is only a tail beyond about $4\,k_1$, which holds 0.2% of the free energy at 64^3 and 0.4% at 256^3.

The Alfvénic fields do cascade along the field lines, and g_0 does not follow them. Figure 5(b) compares the spectra of the whole velocity $\boldsymbol{u}_\perp$ and of g_0 along the same lines. Beyond $4\,k_1$ the velocity holds 1.5%, 2.7% and 3.2% of its energy at 64^3, 96^3 and 128^3, a share that grows as the inertial range lengthens, while g_0 holds 0.11%, 0.14% and 0.18% of its variance there, 14 to 19 times less; its tail grows with the resolution as well, but from an amplitude more than ten times smaller. A passive field that inherited the parallel cascade of the Alfvénic fields would have a spectrum along the lines like that of the velocity.

Filtered into bands of $k_\perp$, g_0 seems to gain field-aligned structure at small perpendicular scales: its κ_1 grows from about $1.0\,k_1$ in the band $k_\perp \le 2k_1$ to 2.0–2.1 k_1 in the band $8 \le k_\perp/k_1 \le 16$ at 64^3 and 128^3 (figure 5c). This growth does not come from structure of g_0 along the field. Filtering mixes neighbouring field lines, which separate along the field, so a filtered field shows field-aligned structure that the whole field does not have. To measure this effect we built from each snapshot a copy of g_0 that is frozen to the field lines by construction: it equals g_0 on the planes $z = 0$, $L/4$,

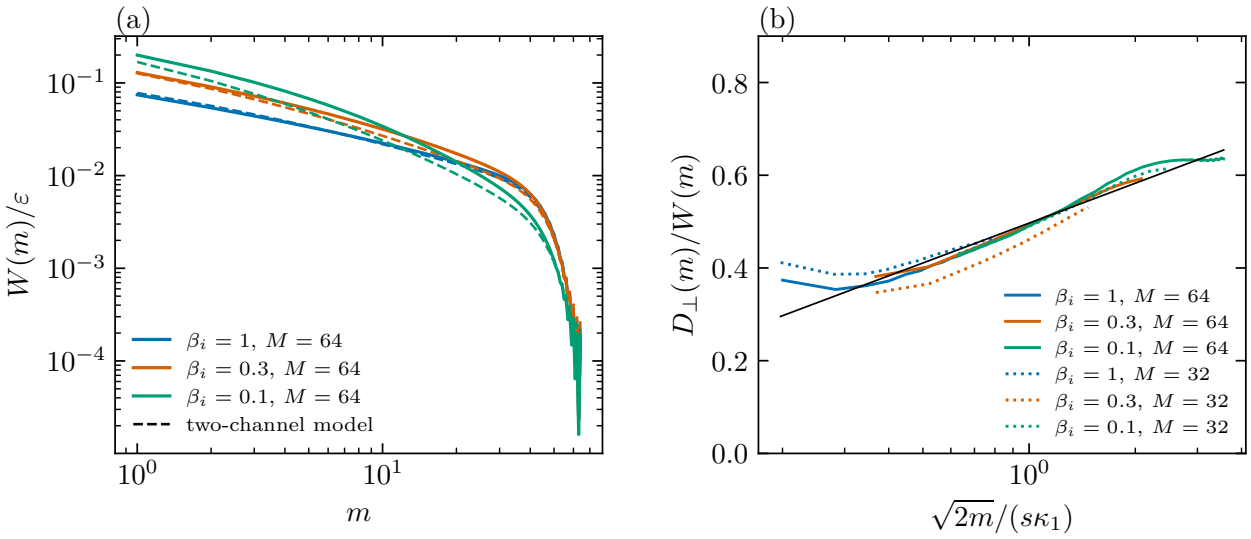


Figure 6: (a) $W(m)/\varepsilon$ at $\beta_i = 1$, 0.3 and 0.1 ($M = 64$) (solid) and the two-channel model with the traced $P(\kappa)$ of the injected pattern and the measured T of each run (dashed). (b) The perpendicular sink $D_\perp(m)/W(m)$ against the time $\sqrt{2m}/(s\kappa_1)$ since injection of the free energy at m, with κ_1 the mean field-aligned wavenumber of the injected pattern, for $1 \le m \le M/2$; the runs at $M = 64$ (solid) reach older free energy than those at $M = 32$ (dotted); the thin line is the joint fit $0.50 + 0.12\ln[\sqrt{2m}/(s\kappa_1)]$.

$L/2$ and $3L/4$ and is carried along the field lines between them, so that along every line it varies only on scales longer than $L/4$ (appendix D.4). Band by band, the field-aligned wavenumber of the frozen copy is that of g_0 to within 4% at both resolutions, whereas that of Φ grows to $3.1\,k_1$. The growth of the band-resolved wavenumber of g_0 therefore comes from the filtering. Over the whole field, g_0 has 0.11–0.18% of its variance beyond $4\,k_1$, against 0.05–0.07% for its frozen copy. The band-resolved wavenumber of a passive field therefore measures the filtering as much as the field, and the spectra of the whole fields, figure 5(a,b), are the test of a parallel cascade.

7.4 The break

At $\beta_i = 1$ the break lies far beyond the highest moment we keep, $m = M = 32$: $m_* = (s\kappa T)^2/2$ is 103 for $\kappa = k_1$ and 185 for the rms field-aligned wavenumber, and $c_T^2\beta_i/(2\chi^2) = 127$ with $c_T = T/\tau_{\rm nl}$. Free energy injected at $m = 0$ reaches $m = 16$ after about $0.8\,\tau_A$ and $m = 32$ after $1.1\,\tau_A$, against $T = 2.3\,\tau_A$, and the hypercollisions remove 57–61% of the injected free energy (the range over the four blocks). The exponential factor in (42) therefore reduces W by only 29% at $m = 16$ and 39% at $m = 32$, and at $\beta_i = 1$ the test of the shape confirms that there is no deficit, not that the sink produces a break.

Lowering β_i at fixed Alfvénic forcing moves the break into the resolved range, as prediction (iii) states. We ran $\beta_i = 0.3$ and 0.1, each at $M = 32$ and $M = 64$, and $\beta_i = 1$ also at $M = 64$; the branch is $\Lambda^+ = 6.2206$ and 19.198. With T and $P(\kappa)$ from $\beta_i = 1$, (45) gives $m_* \approx 31$–56 at $\beta_i = 0.3$ and $m_* \approx 10$–19 at $\beta_i = 0.1$. At $\beta_i = 0.1$ the break with the measured T, $m_* \approx 6$–10, lies well inside the resolved range, and $W(m)$ falls below $m^{-1/2}$ from small m (figure 6a). The continuum formula (42) with $\kappa = k_1$ and the measured T follows the measured spectrum to about 10% up to $m \approx 16$, where the hypercollisions take over at $M = 32$; at $M = 64$ the spectrum continues to fall steeply between $m \approx 15$ and 40. At $\beta_i = 0.3$ the spectrum bends at correspondingly larger m.

Table 3: The runs of the m_* scan, on 64^3 points. η is the hyper-resistivity and the window is as in table 2; $n_e = (\sqrt{M} - 2)/\ell_*$, with ℓ_* from the fit over the whole collision-free range $2 \le m \le M/2$, is the number of decay lengths that fit in it; $m_* = \ell_*^2/2$ from the fit over $2 \le \sqrt{2m} \le 2 + 1.5\,\ell_*$, with the standard error of the fits to the four block averages; $1/T$ is the median of $D_\perp(m)/W(m)$ over $2 \le m \le 16$ and κ_1 the mean field-aligned wavenumber of the injected pattern.

β_i	χ	M	η	window	n_e	m_*	$(s\kappa_1 T)^2/2$	$T/\tau_{\rm nl}$
0.05	0.96	128	300	20–180	3.45	3.7 ± 0.3	3.3	10.9
0.1	0.96	64	300	16–66	1.69	6.6 ± 0.4	6.9	11.0
0.3	0.94	192	300	28.5–66	1.73	24.2 ± 0.6	31.4	13.6
0.05	0.75	128	219	20–200	2.65	7.0 ± 0.1	6.6	11.9
0.05	0.56	256	156	30–300	2.81	14.9 ± 0.6	16.6	13.9
0.05	0.40	512	33	30–330	1.87	60.5 ± 1.1	167	25.9

The decay beyond the break has the predicted rate. We fit $W(m)\sqrt{2m} \propto \exp(-\sqrt{2m}/\ell_*)$ over $2 \le m \le M/2$, below the collisional range. The fitted ℓ_* is 16.8, 8.3 and 3.8 at $\beta_i = 1$, 0.3 and 0.1 ($M = 32$), and 15.7, 7.2 and 3.6 at $M = 64$, against $s\kappa_1 T = 16.4$, 8.0, 3.8, 16.8, 7.6 and 3.7 from the measured T and the injected κ_1: the measured decay rate in $\sqrt{2m}$ is the predicted one to within 7% in every run. At $\beta_i = 1$ the fit range spans less than half a decay length ($\sqrt{2m} \le \sqrt{M} = 5.7$ and 8 against $\ell_* \approx 16$), so the break $m_* = \ell_*^2/2 \approx 140$ there extrapolates the fit well beyond $m = M$; at $\beta_i = 0.1$ and $M = 64$ the range spans 1.7 decay lengths and $m_* \approx 7$.

Of the runs in table 2, only those at $\beta_i = 0.1$ place the break well inside the collision-free range $m \le M/2$. To measure m_* against β_i and χ we made the runs of table 3, with M large enough in each that the break lies below $M/2$. At $\chi \approx 0.95$ we ran $\beta_i = 0.05$ and 0.3, and we use the run at $\beta_i = 0.1$ and $M = 64$ of table 2. At $\beta_i = 0.05$ we lowered χ to 0.75, 0.56 and 0.40 by lowering the Alfvénic injection rate ε_A from 0.3 to 0.1, 0.031 and 0.0035. As χ decreases the hyper-resistive dissipation of E_A moves to larger scales unless η decreases with it, so we lowered η in proportion to χ, to 219 and 156, and at $\chi = 0.40$ set $\eta = 33$ and ε_A by short runs at $M = 2$; the median shell of the hyper-resistive dissipation of E_A is then $k_\perp/k_1 = 9$–11 in every run. We fit over a range set by each run's own decay, from $m = 2$ to $\sqrt{2m} = 2 + 1.5\,\ell_*$; the collision-free range holds 1.7 to 3.5 decay lengths, and at the top of the fit range the hypercollisional rate $2\nu_m$ is at most 13% of $1/T$. The model value is $(s\kappa_1 T)^2/2$, with $1/T$ the median rate over $2 \le m \le 16$ of each run and κ_1 from lines traced on its snapshots.

At $\chi \approx 0.95$ the break grows from $m_* = 3.7$ at $\beta_i = 0.05$ to 6.6 and 24.2 at $\beta_i = 0.1$ and 0.3 (figure 7a). A weighted fit gives $m_* \propto \beta_i^{1.09\pm0.04}$, close to the $m_* \propto \beta_i$ of (49). The model values follow $\beta_i^{1.31\pm0.04}$. They agree with the measured m_* to within 12% at $\beta_i \le 0.1$ and lie 30% above it at $\beta_i = 0.3$, where $T/\tau_{\rm nl}$ is larger, 13.6 against 10.9 and 11.0, because the moments $2 \le m \le 16$ that set T hold younger free energy at larger β_i (section 7.5).

At $\beta_i = 0.05$ the break grows from $m_* = 3.7$ at $\chi = 0.96$ to 7.0, 14.9 and 60.5 at $\chi = 0.75$, 0.56 and 0.40 (figure 7b). Between neighbouring runs the local exponent $-\mathrm{d}\ln m_*/\mathrm{d}\ln\chi$ is 2.5, 2.7 and 4.1. Down to $\chi \approx 0.56$ the break therefore grows somewhat faster than the χ^{-2} of (49), and below it the growth steepens towards χ^{-4}; no single power law describes the four runs. The model value follows the measured m_* to within 12% for $\chi \ge 0.56$, but at $\chi = 0.40$ it is 167, 2.8 times the measured value. Both departures come from the perpendicular sink (section 7.5).

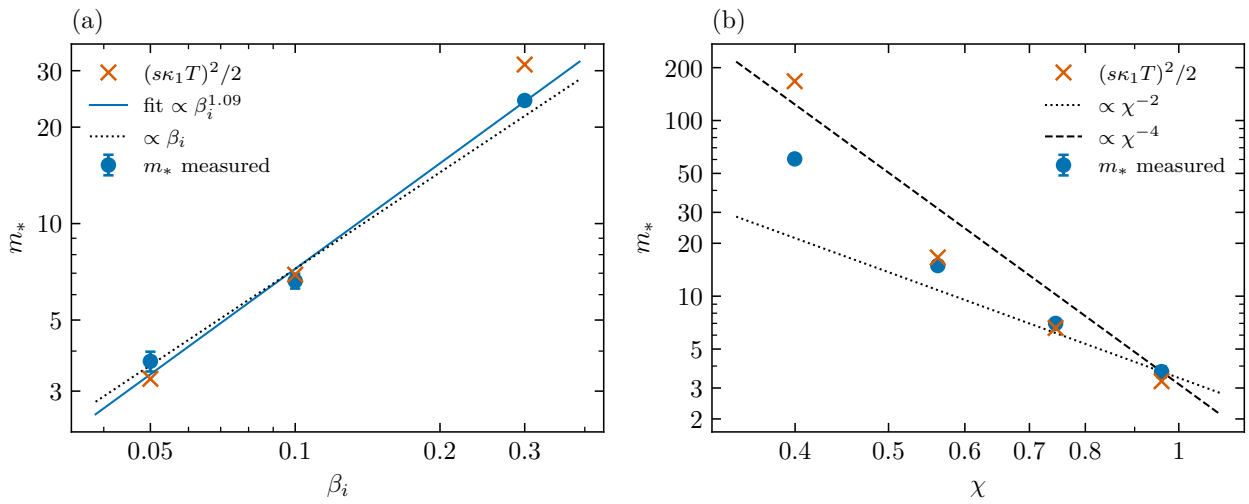


Figure 7: The break $m_* = \ell_*^2/2$ from the fit to $W(m)$ (circles, with the standard error of the four block fits) and the model value $(s\kappa_1 T)^2/2$ (crosses). (a) Against β_i at $\chi \approx 0.95$, with a weighted power-law fit (solid) and $m_* \propto \beta_i$ (dotted). (b) Against the outer-scale χ at $\beta_i = 0.05$, with χ^{-2} (dotted) and χ^{-4} (dashed) through the run at $\chi = 0.96$.

7.5 The age of the free energy

The perpendicular sink is not uniform in m but depends on the age of the free energy, in the same way at every β_i. In the runs at 64^3 and $\chi \approx 1$, $D_\perp(m)/W(m)$ rises with m, from 0.33–0.43 at $m \le 1$ to 0.46, 0.53 and 0.62 at $m = 16$ for $\beta_i = 1$, 0.3 and 0.1, and the rise is steeper at lower β_i, where free energy reaches a given m later. Plotted against the time $\vartheta = \sqrt{2m}/(s\kappa_1)$ since injection, the sinks of these runs fall on one curve (figure 6b), with an rms scatter of 4.5% about a joint fit $D_\perp/W = 0.50 + 0.12 \ln \vartheta$, against 12.6% when plotted against m. $D_\perp/W$ is about 0.4 at the smallest ages and levels off at about 0.63 for ages beyond about $2.5\,\tau_A$. This is what is expected of a passive scalar whose variance reaches the dissipation scale only after the cascade has had time to act (section 6.4). This favours the age over the streaming of the slices as the cause: at a fixed age the streaming speed of the slices that dominate moment m, $s\sqrt{2m}$, scales as s^2 and differs by a factor of ten between $\beta_i = 1$ and 0.1, yet the sinks fall on one curve.

A single rate $1/T$, the median over $2 \le m \le 16$, therefore removes too much free energy at small m. At $\beta_i = 1$ the measured $W(m)$ is within 7% of the model with the traced $P(\kappa)$ for $1 \le m \le M/2$, at both M. At lower β_i it lies above the model at every m, by up to 21% and 39% at $\beta_i = 0.3$ and 0.1 at $M = 32$ and by up to 20% and 42% at $M = 64$, and the field-aligned flux lies above the model as well (figure 6a). The median rate $1/T$ rises from 0.44 to 0.59 between $\beta_i = 1$ and 0.1 for the same reason: at low β_i the moments $2 \le m \le 16$ hold older free energy.

To test this directly we let each injection evolve by the linear hierarchy at its field-aligned wavenumber, with the run's hypercollisions, and remove it at a rate $\gamma(\vartheta)$ that depends only on its age, the same at every m. With $S(\vartheta) = \exp(-\int_0^\vartheta \gamma)$ the steady state is

$$W(m) = \varepsilon \sum_\kappa P(\kappa) \int_0^\infty E_m(\vartheta;\kappa)\, S(\vartheta)\, \mathrm{d}\vartheta, \tag{51}$$

with $E_m(\vartheta;\kappa)$ the free energy at moment m of a unit injection at age ϑ, and $D_\perp(m)$ is the same integral with γS in place of S; for $\gamma = 1/T$ this is the uniform-sink model. Fitting $\gamma(\vartheta) = \gamma_0 + (\gamma_1 - \gamma_0)(1 - e^{-\vartheta/\vartheta_c})$ to $D_\perp(m)/W(m)$ at $\beta_i = 0.1$ ($M = 64$) gives $\gamma_0 = 0.03$, $\gamma_1 = 0.67$

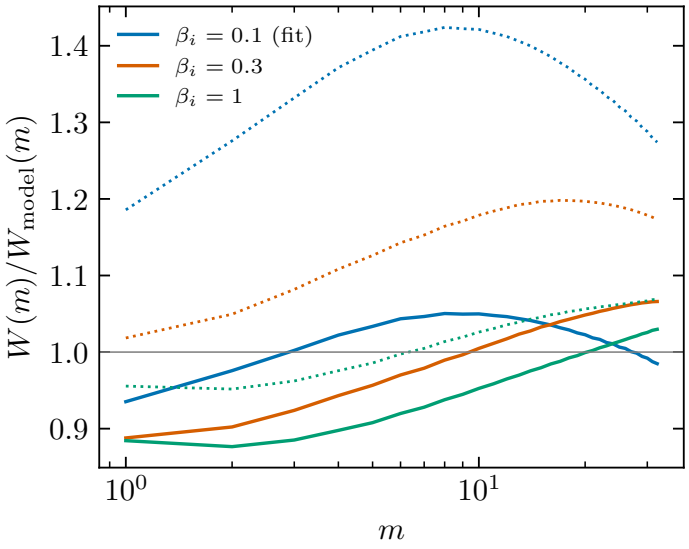


Figure 8: $W(m)$ divided by the two-channel model with the age-dependent sink $\gamma(\vartheta)$ fitted at $\beta_i = 0.1$ (solid) and with the uniform sink $1/T$ of each run (dotted), for the runs at $M = 64$.

and $\vartheta_c = 0.76\,\tau_A$, about five outer eddy times: fresh free energy is hardly removed, and the sink switches on as the cascade carries it to the dissipation scale; $D_\perp/W$ exceeds γ_0 even at small m because each moment also holds older free energy at small κ. With this $\gamma(\vartheta)$ and no parameter fitted to them, the model gives $D_\perp(m)/W(m)$ at $\beta_i = 0.3$ and 1 to within 5% for $m \geq 2$ and $m \geq 4$, across a factor of ten in β_i. It gives $W(m)$ to within 7% for $m \geq 8$ in all three runs, and lies above the measured W at the smallest m, by 7% at $m = 1$ in the run used for the fit and by 8–14% for $m \leq 3$ in the other two (figure 8).

The same property explains the departures of m_* from χ^{-2} (section 7.4). With the age and the rate both measured in units of τ_{nl}, the sinks of the runs at $\chi \geq 0.56$ agree to within about 10% for $\vartheta \geq 12\,\tau_{\mathrm{nl}}$: the rate switches on over the first 10–15 τ_{nl} and levels off at 0.087–0.094/τ_{nl}, slightly lower at smaller χ. The range $2 \leq m \leq 16$ that defines T covers the same ages in units of τ_A in every run at $\beta_i = 0.05$, but as χ decreases and τ_{nl} grows it holds free energy that is younger in units of τ_{nl}, from 8.5–24 τ_{nl} at $\chi = 0.96$ to 5–14 τ_{nl} at $\chi = 0.56$. The sink there is weaker, so T/τ_{nl} rises from 10.9 to 13.9 and m_* grows faster than χ^{-2}. At $\chi = 0.40$ the moments $2 \leq m \leq 16$ hold free energy of age 3–8 τ_{nl}, and the sink itself changes: it switches on as at larger χ, reaches 0.069/τ_{nl} between $\vartheta = 12$ and 16 τ_{nl} and falls to 0.055/τ_{nl} at 30 τ_{nl}, where the runs at larger χ are at 0.087–0.094/τ_{nl}. The cascade then removes the free energy more slowly per outer eddy time, as expected once the outer-scale turbulence is weak and its cascade time exceeds τ_{nl}. Over $1 \leq m \leq M/2$ the rate $D_\perp/W$ varies by a factor of 10 at $\chi = 0.40$, against 1.4, 1.7 and 2.5 at $\chi = 0.96$, 0.75 and 0.56, so the median over $2 \leq m \leq 16$ no longer represents the rate over the range $m \leq 171$ on which the break is measured, and $(s\kappa_1 T)^2/2$ overestimates m_*. With the rate averaged over each run's fit range instead, uniformly in $\sqrt{2m}$ as the fit weights it, $s\kappa_1 T$ agrees with the fitted ℓ_* to within 9% at $\chi = 0.40$ and to within 5% in every other run of table 3. The decay beyond the break is therefore still set by phase mixing at κ_1 against the perpendicular sink, as in (50). What departs from (49) at small χ is the assumption that the sink rate is a fixed multiple of $1/\tau_{\mathrm{nl}}$.

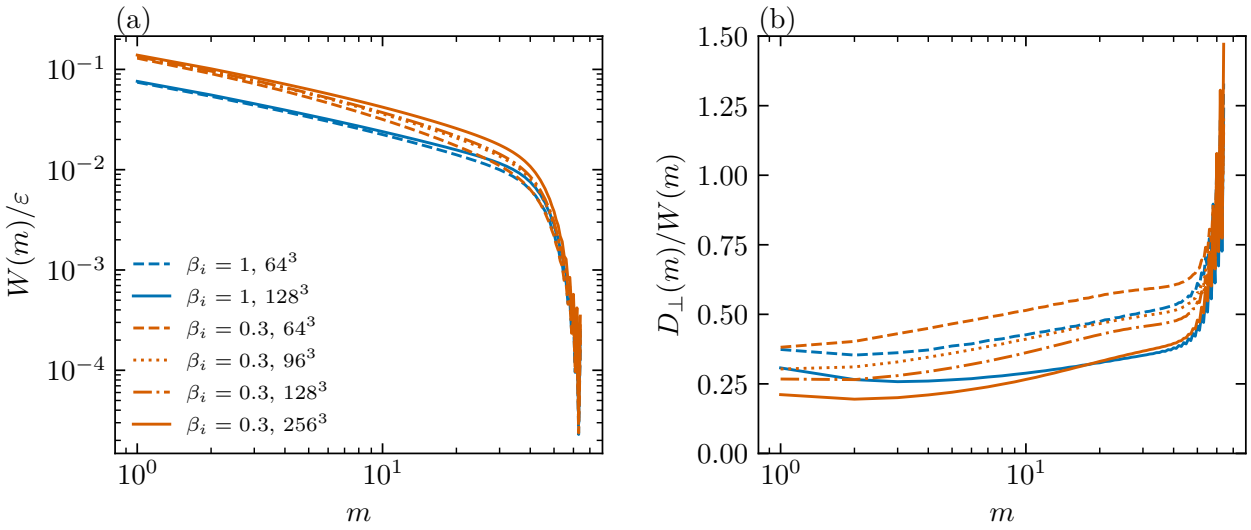


Figure 9: Resolution in $k_\perp$ at $M = 64$, with the hyper-resistivity normalised to the grid: $\beta_i = 1$ (blue) at 64^3 (dashed) and 128^3 (solid), and $\beta_i = 0.3$ (orange) at 64^3 (dashed), 96^3 (dotted), 128^3 (dash-dotted) and 256^3 (solid). (a) $W(m)/\varepsilon$. (b) $D_\perp(m)/W(m)$.

7.6 Dependence on resolution

Doubling M from 32 to 64 leaves $W(m)$ unchanged to within 5% for $1 \le m \le 10$ and changes $1/T$ by at most 6% at every β_i. What changes is the share of the free energy that reaches the hypercollisions. When the collisional range moves from $m > 16$ to $m > 32$ it falls from 59% to 50% at $\beta_i = 1$, from 43% to 27% at $\beta_i = 0.3$ and from 20% to 10% at $\beta_i = 0.1$, and the model with the traced $P(\kappa)$ gives the shares to within 0.06 at $M = 32$ and 0.02 at $M = 64$. The collisional share of the dissipation is therefore not a converged quantity at $M = 32$: free energy that the hypercollisions remove at a low cutoff is removed at small perpendicular scales when the cutoff is raised.

The perpendicular resolution changes the sink. We repeated the runs at $M = 64$ on 96^3, 128^3 and 256^3 grid points at $\beta_i = 0.3$ and on 128^3 at $\beta_i = 1$, with the hyper-resistivity normalised to the grid, so that the inertial range lengthens with N: the slope of E_A fitted over the middle 60% of the resolved shells goes from -4.1 at 64^3 to -2.9 at 256^3. The outer scale is unchanged, with $\tau_{\rm nl} = 0.150$–0.158 and $\chi = 0.92$–0.97 in all six runs. $D_\perp(m)/W(m)$ falls at every m, by 18–35% with each doubling of N (figure 9), and $T/\tau_{\rm nl}$ rises from 13.2 at 64^3 to 16.4, 18.4 and 24.6 at $\beta_i = 0.3$, and from 15.4 to 22.6 at $\beta_i = 1$. At $\beta_i = 0.3$, T grows by 5–6 outer eddy times with each doubling, with no sign of levelling off up to 256^3. The break moves out with it, from $m_* \approx 26$ at 64^3 to 86 at 256^3 (from the fit over $2 \le m \le M/2$, which at 256^3 extrapolates beyond $m = M$), the share of the free energy removed by hypercollisions rises from 27% to 46%, and $W(m)$ at $m \le 16$ rises by up to 43%.

The model follows these changes. At 128^3 and 256^3, with the T and $P(\kappa)$ measured there, it describes $W(m)$ to within -9% and $+13\%$ and the collisional shares to within 0.02, and the field-aligned flux lies 3–15% above it, with no deficit at large m. The measured decay rate in $\sqrt{2m}$ is within 5% of $1/(s\kappa_1 T)$, except at $\beta_i = 0.3$ at 128^3 and 256^3, where it is 14% and 17% faster.

At the resolutions we reach, then, the free energy keeps its injected field-aligned wavenumber (section 7.3), but the rate of the perpendicular sink is not independent of the dissipation scale, and prediction (iv) is not confirmed. Either the dissipation range of our runs reaches almost to the

forcing scales and makes the sink artificially fast, in which case T levels off once the inertial range is long enough, or the perpendicular cascade of the slices has no dissipative anomaly, in which case T, and with it m_*, keeps growing as the dissipation weakens. Up to 256^3 we see no sign of levelling off, but our runs cannot exclude it at higher resolution.

8 Discussion

Two results of this paper are exact. The compressive sector of KRMHD is a continuum of passive slices labelled by phase velocity, for each branch and in truncated hierarchies, for any smooth Alfvénic fields and without collisions (theorems 1 and 2), so the nonlinearity moves no free energy between phase velocities. With an ideal induction equation, smooth fields and no spatial dissipation of the compressive distribution, every volume-integrated single-point statistic, including $W(m)$ and the field-aligned Hermite flux, evolves as in linear phase mixing along the initial field lines (theorem 3), so such dynamics produces no stochastic echoes in $W(m)$.

The two-channel model extends these results to turbulence, where they do not apply directly, by treating spatial dissipation as a sink that removes free energy from every Hermite moment at the passive-scalar cascade rate. In the pilot simulation at $\beta_i = 1$ and $\chi \approx 1$ the model holds to the accuracy of the measurement. The perpendicular dissipation is close to an m-independent sink, and $W(m)$ and the field-aligned flux are those of phase mixing at the injected field-aligned wavenumber with that sink, with no deficit below the collisional range. The free energy keeps its injected field-aligned wavenumber: at $\beta_i = 0.3$ its field-aligned spectrum is the same from 64^3 to 256^3 grid points, apart from a tail holding less than 0.5% of it, so in our runs the relabelling of field lines that Meyrand et al. (2019) invoke does not give g the parallel structure of the Alfvénic fields.

Our spectrum at $\beta_i = 1$ differs from that of Meyrand et al. (2019), who found $W(m)$ somewhat steeper than m^{-1} at the same β_i and a net flux to high m much smaller than the injection rate. Their set-up differs from ours in four ways that matter here. They forced the compressive fluctuations at $m = 1$, the parallel flow, on modes with $|k_x|, |k_y|, |k_z| \leq 2$, where we force g_0 at $|k_z| = k_1$; they damped g with $\mu k_\perp^8$ and νm^6, where we use a fourth power of $k_\perp$ normalised to the grid and $\nu(m/M)^6$; they ran at 256^3 to 512^3 with $M = 32$ to 128, against our runs at $\beta_i = 1$ on 64^3 and 128^3 with $M = 32$ and 64; and their Hermite flux is resolved in $k_\perp$ shells, which only in its sum over shells is a pure flux in m. In our model each of the first three differences raises the break rather than lowering it: forcing up to $|k_z| = 2k_1$ raises the injected κ, and a longer inertial range with sharper dissipation lengthens T, as our runs at 96^3 to 256^3 show, and $m_* = (s\kappa T)^2/2$ grows with both. The model therefore does not account for their steeper spectrum by these differences alone. We tested the forcing directly (section 7.2). With g_1 forced on $|k_z| = k_1$ and $2k_1$ at 64^3, $W(m)$ steepens at small m, from $m^{-0.54}$ to $m^{-0.66}$ over $2 \leq m \leq 8$, keeps a slope of about -0.7 beyond, and is reproduced by the model with the injected $P(\kappa)$ to within 3.5%, with no return flux; at $m = 16$ the net flux still carries 75% of the injection. The forcing at $m = 1$ therefore does not produce their spectrum at our resolution. That leaves the different dissipation of g and effects that appear only at their higher resolution, such as echoes that our runs do not resolve; in our runs a higher resolution makes the spectrum shallower, not steeper. Their evidence that g inherits the parallel cascade of the Alfvénic fields comes from tracing the perturbed field lines (their figure 3): the parallel spectra of the density and field-strength fluctuations along the lines track that of the Alfvénic velocity, and the parallel coherence length of the compressive fluctuations grows with their perpendicular one as that of the Alfvénic fluctuations does. In our runs the first does not hold: along the same lines the spectrum of the density moment stays at the injected wavenumbers, with 0.18% of its variance beyond $4\,k_1$ at 128^3 against 3.2% of the energy of the velocity (section 7.3).

The second compares the fields at a given perpendicular separation, a scale-by-scale measure like the band-resolved wavenumber of section 7.3, which gives a copy of g_0 frozen to the field lines by construction the same growth with $k_\perp$ as g_0 itself. Running our diagnostics, including the frozen copy, on their set-up would show whether the difference lies in the set-up or in the measurement. Meyrand et al. (2019) did not split the flux by the sign of the parallel wavenumber, which, as they note, would have to be taken along the perturbed field. A split by the sign of k_z, the Eulerian version of that decomposition, reports in our runs a backward flux of about 12% of the forward flux at 64^3 and about 30% at 256^3, where the field-aligned flux has none.

At fixed χ the model predicts that m_* grows with β_i, as $\beta_i T^2$. At $\beta_i = 1$ the break lies beyond the highest Hermite moment we keep, $m = M$; at $\beta_i = 0.1$ it lies well below it and $W(m)$ steepens from small m, as predicted, with the same Alfvénic turbulence in every run. The trend that Kawazura et al. (2019, 2020) found in gyrokinetic simulations, $m^{-1/2}$ at high β_i and close to m^{-1} at $\beta_i = 0.1$, is the trend the model gives without any echo, because m_* falls with β_i. Fitted to the measured spectra, the decay rate in $\sqrt{2m}$ is the predicted $1/(s\kappa_1 T)$ to within 7% at every β_i and both M at 64^3. With M raised until the break is resolved, m_* grows as $\beta_i^{1.09\pm0.04}$ at $\chi \approx 0.95$ for $0.05 \le \beta_i \le 0.3$, close to the $m_* \propto \beta_i$ of (49). Extending this measurement to $\beta_i = 1$, where resolving the break needs M of several hundred, is ongoing work. The comparison also showed where the model is too simple: the perpendicular sink is not uniform in m but grows with the age of the free energy, and with that refinement (50) the sinks of the runs at 64^3 and $\chi \approx 1$ collapse onto one function of age. Against χ the estimate $m_* \sim \beta_i/\chi^2$ holds only near $\chi \approx 1$. At $\beta_i = 0.05$ the local exponent of m_* in χ is -2.5 and -2.7 down to $\chi \approx 0.56$ and -4.1 between 0.56 and 0.40. In units of the outer eddy time the sink is close to the same function of age at $\chi \ge 0.56$ and weaker at $\chi = 0.40$, as expected when the outer-scale turbulence becomes weak. With the rate averaged over the range where the break is fitted, the decay length is still $s\kappa_1 T$ to within 9%, so what fails at small χ is the assumption $T \propto \tau_{\mathrm{nl}}$, not the competition between phase mixing and the sink. The model also predicts that the share of the free energy dissipated by collisions falls as the collisional cutoff m_c rises, as $\exp(-\sqrt{2m_c}/(s\kappa T))$; the runs at $M = 32$ and $M = 64$ show this fall, and with the traced $P(\kappa)$ the model gives the shares to within 0.06. If this holds as $m_c \to \infty$, the free energy that phase mixes is in the end removed at small perpendicular scales, and the fraction dissipated by collisions in velocity space at fixed β_i and χ is set by where the collisions act; in a real plasma the perpendicular cascade of the slices stops at the ion gyroscale, beyond the reach of KRMHD.

The main open question is the limit of high resolution. The model needs a perpendicular cascade whose rate does not depend on the dissipation coefficients and field lines that stay frozen for the free energy that phase mixes. Drivas and Eyink (2017) and Eyink (2018) give reasons to expect that the first requires the failure of the second at small enough scales (section 6.4). Our runs do not reach a cascade rate independent of the dissipation. Up to 256^3 grid points T grows by about the same amount with each doubling of the resolution, from 13 to 25 outer eddy times at $\beta_i = 0.3$ (section 7.6), while the field lines stay frozen for the free energy that phase mixes, whose field-aligned spectrum does not change (section 7.3). In our runs the two therefore go together as that argument suggests: the field lines stay frozen and the sink shows no anomaly. If T levels off at higher resolution, the argument implies that the field lines stop being frozen at the scales where the sink acts, and the model then holds only if the free energy that phase mixes sits at larger scales (section 6.4). With the T and κ measured at each resolution the model still describes $W(m)$ to within -9% and $+13\%$, and there is still no return flux. The model therefore describes the spectrum at each resolution with the T measured there, but in our runs T, and with it m_*, has no value independent of the dissipation. Whether T converges once the inertial range is long enough, or keeps growing so that the perpendicular cascade loses ground to phase mixing as the dissipation weakens, needs higher resolution than we reach.

The results hold for $k_\perp \rho_i \to 0$. At finite $k_\perp \rho_i$ the Landau coupling depends on the wavenumber, the advection depends on $v_\perp$ and the compressive fluctuations act back on the Alfvénic ones (Chandran et al., 2026), so the slices no longer decouple. Velocity-dependent collisions are the only terms in KRMHD that exchange free energy between slices (18).

A The two compressive branches

We derive the branches of section 2.2 following S09 (§6.2) and CMM (appendix B), with $\delta_n \equiv \delta n/n$, $\delta_B \equiv \delta B/B$ and $\langle\cdot\rangle$ the volume average. We checked every identity symbolically, and the moments and splits also by quadrature over $(v_\perp, v_\parallel)$ for random distribution functions.

A.1 Laguerre projection and the branches

With $x = v_\perp^2/v_{\rm th}^2$, CMM (eq. B8) and S09 (eq. 205) write

$$g = \frac{n_i}{\pi v_{\rm th}^2}\, e^{-x} \sum_{l\geq 0} L_l(x)\, G_l(v_\parallel), \qquad F_0 = \frac{n_i}{\pi v_{\rm th}^2}\, e^{-x} F_M(v_\parallel),$$

with L_l orthonormal with weight e^{-x} on $(0,\infty)$. Since $x = L_0 - L_1$ and $x^k \perp L_l$ for $l > k$, the source term of (3) projects only on L_0 and L_1, and (3) and (4) become

$$\frac{\mathrm{d}G_0}{\mathrm{d}t} + v_\parallel \nabla_\parallel \Big[G_0 + F_M\Big(\frac{Z}{\tau}\delta_n + \delta_B\Big)\Big] = 0, \qquad \frac{\mathrm{d}G_1}{\mathrm{d}t} + v_\parallel \nabla_\parallel (G_1 - F_M \delta_B) = 0,$$
$$\frac{\mathrm{d}G_l}{\mathrm{d}t} + v_\parallel \nabla_\parallel G_l = 0 \quad (l \geq 2),$$
$$\int G_0 \,\mathrm{d}v_\parallel = \delta_n - \delta_B, \qquad \int (G_0 - G_1)\,\mathrm{d}v_\parallel = -\frac{Z}{\tau}\delta_n - 2\Big(1 + \frac{1}{\beta_i}\Big)\delta_B.$$

With $D = Z/\tau + 2(1+\beta_i^{-1})$, the distributions of CMM's eqs (B2)–(B3) (S09 eqs 176–177),

$$G_n = -\frac{1}{D}\Big[(G_0 - G_1) - 2\Big(1 + \frac{1}{\beta_i}\Big)G_0\Big], \qquad G_B = -\frac{1}{D}\Big[(G_0 - G_1) + \frac{Z}{\tau}G_0\Big],$$

have $\int G_n \,\mathrm{d}v_\parallel = \delta_n$ and $\int G_B \,\mathrm{d}v_\parallel = \delta_B$, invert to

$$G_0 = G_n - G_B, \qquad G_1 = \Big(1 + \frac{Z}{\tau}\Big)G_0 + D\, G_B, \tag{A1}$$

and obey S09's eqs (179)–(180),

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} G_n \\ G_B \end{pmatrix} + v_\parallel \nabla_\parallel \begin{pmatrix} G_n \\ G_B \end{pmatrix} = v_\parallel F_M \,\nabla_\parallel K \begin{pmatrix} \delta_n \\ \delta_B \end{pmatrix}, \qquad K = \frac{1}{D}\begin{pmatrix} -\frac{Z}{\tau}\big(1 + \frac{2}{\beta_i}\big) & -\frac{2}{\beta_i} \\ \frac{Z}{\tau}\big(1 + \frac{Z}{\tau}\big) & 2 + \frac{Z}{\tau} \end{pmatrix}.$$

If $\ell K = \Lambda^{-1}\ell$, then $G = \ell \cdot (G_n, G_B)$ has $\int G \,\mathrm{d}v_\parallel = \ell \cdot (\delta_n, \delta_B)$ and obeys (5). The two left eigenvectors give CMM's branches (B5)–(B6) (S09 eqs 183–184),

$$G^+ = G_B + \frac{1 + Z/\tau}{\sigma}\, G_n, \qquad G^- = G_n + \frac{2\tau}{\sigma Z \beta_i}\, G_B, \qquad \sigma = 1 + \frac{\tau}{Z} + \frac{1}{\beta_i} + \Delta,$$

and the eigenvalues are $1/\Lambda^\pm$ with $\Lambda^\pm$ and Δ of (6).

A.2 Moments

The map $(G_n, G_B) \mapsto (G^+, G^-)$ has determinant $-2\Delta/\sigma \neq 0$ and inverse

$$G_n = \frac{\sigma G^- - (2\tau/Z\beta_i)\, G^+}{2\Delta}, \qquad G_B = \frac{\sigma G^+ - (1 + Z/\tau)\, G^-}{2\Delta}, \tag{A2}$$

so that, with $M_0^\pm = \int G^\pm \, \mathrm{d}v_\parallel$,

$$\delta_n = \frac{\sigma M_0^- - (2\tau/Z\beta_i)\, M_0^+}{2\Delta}, \qquad \delta_B = \frac{\sigma M_0^+ - (1 + Z/\tau)\, M_0^-}{2\Delta}. \tag{A3}$$

These are one half of the right-hand sides of S09's eqs (202)–(203), as CMM found (footnote to their eq. B15). The parallel flow and pressure perturbations of the non-Alfvénic part $g_i = g + x\,\delta_B F_0$ of the ion distribution (CMM eq. 3.26) are

$$u_\parallel = \int v_\parallel G_0 \, \mathrm{d}v_\parallel, \qquad \frac{\delta p_\perp}{p} = -\frac{Z}{\tau}\delta_n - \frac{2}{\beta_i}\delta_B, \qquad \frac{\delta p_\parallel}{p} = \frac{2}{v_{\mathrm{th}}^2}\int v_\parallel^2 G_0 \, \mathrm{d}v_\parallel + \delta_B,$$

with $p = n_i T_i$. By (A1) and (A2) these are linear in $G^\pm$, and the G_l with $l \geq 2$ enter only moments $\int \mathrm{d}^3 v \, v_\parallel^j v_\perp^{2k} g_i$ with $k \geq 2$.

A.3 Free energy

With $x = L_0 - L_1$, the orthonormality of the L_l and the field relations, CMM's compressive free energy (their eq. 3.31) becomes

$$\begin{aligned} W_{\mathrm{compr}} &= \frac{n_i T_i}{2}\Big\langle \frac{Z}{\tau}\delta_n^2 + \frac{2}{\beta_i}\delta_B^2 + \frac{1}{n_i}\int \mathrm{d}^3 v \, \frac{g_i^2}{F_0}\Big\rangle \\ &= \frac{n_i T_i}{2}\Big\langle \int \frac{G_0^2 + G_1^2}{F_M}\,\mathrm{d}v_\parallel + \frac{Z}{\tau}\delta_n^2 - \frac{2Z}{\tau}\delta_n\delta_B - 2\Big(1 + \frac{1}{\beta_i}\Big)\delta_B^2 + \sum_{l\geq 2}\int \frac{G_l^2}{F_M}\,\mathrm{d}v_\parallel\Big\rangle. \end{aligned}$$

The part with $l = 0, 1$ splits between the branches by the identities

$$G_0^2 + G_1^2 = c_+ (G^+)^2 + c_- (G^-)^2, \qquad \frac{Z}{\tau}\delta_n^2 - \frac{2Z}{\tau}\delta_n\delta_B - 2\Big(1 + \frac{1}{\beta_i}\Big)\delta_B^2 = -c_+\frac{(M_0^+)^2}{\Lambda^+} - c_-\frac{(M_0^-)^2}{\Lambda^-}, \tag{A4}$$

the first of which holds at each $v_\parallel$, with

$$c_+ = \Big[1 + \frac{1 + \tau/Z}{\Delta}\Big](\Lambda^+)^2 > 0, \qquad c_- = \frac{Z^2}{2\tau^2}\Big[1 + \frac{1}{\Delta\beta_i}\Big](\Lambda^-)^2 > 0. \tag{A5}$$

This gives CMM's eq. (B15) with their eqs (B13)–(B14),

$$W_{\mathrm{compr}} = \tilde{W}_{\mathrm{compr}} + c_+ W_{\mathrm{compr}}^+ + c_- W_{\mathrm{compr}}^-, \qquad W_{\mathrm{compr}}^\pm = \frac{n_i T_i}{2}\Big\langle \int \frac{(G^\pm)^2}{F_M}\,\mathrm{d}v_\parallel - \frac{(M_0^\pm)^2}{\Lambda^\pm}\Big\rangle, \tag{A6}$$

with $\tilde{W}_{\mathrm{compr}}$ the part with $l \geq 2$. Each part is conserved (S09, §§6.2.3 and 6.2.5; CMM eq. B12) and positive definite (section 3).

The coefficients (A5) use Δ of (6) (S09 eq. 204) and are one quarter of those in S09's eq. (213), as CMM state. CMM's eq. (B16) prints this square root as $[(1 + \tau/Z) + \beta_i^{-2}]^{1/2}$, which gives $\sqrt{3}$ instead of $\sqrt{5}$ at $\beta_i = \tau = Z = 1$. Their eqs (B6)–(B7) are correct, so the misprint affects only $c_\pm$ in (B15) and (B17) and our (A6), which then fails. For random distribution functions the relative error in W_{compr} with the printed square root is up to 7.2×10^{-2} over the parameters we tried (2.5×10^{-2} at $\beta_i = \tau = Z = 1$), and below 5×10^{-14} with the square restored.

B The slice transform and the spectral theory of the branch operator

B.1 Definitions and statement

Let $\mathcal{W}$ be the space of real functions $G(u)$ with $\langle G, G\rangle < \infty$, where $\langle f, g\rangle = \int f g\, F_M^{-1}\,\mathrm{d}u$, so that $\mathsf{Q} = F_M\, M_0$ is the orthogonal projector onto F_M. The free-energy product (12) is $[f, g] = \langle f, Pg\rangle$ with $P = I - \mathsf{Q}/\Lambda$, and with $G = M_0[G]\, F_M + G_\perp$, $[G, G] = \langle G_\perp, G_\perp\rangle + p_0\, M_0[G]^2$, which is positive definite, with a norm equivalent to that of $\langle\cdot,\cdot\rangle$, if and only if $\Lambda \notin [0, 1]$. With U the multiplication by u,

$$L = UP, \tag{B1}$$

so $[f, Lg] = \langle Pf, UPg\rangle$ is symmetric in f and g. For $\operatorname{Im} z > 0$,

$$(z - L)^{-1}G = \frac{1}{z-u}\left[G - \frac{uF_M}{\Lambda\,\mathcal{K}(z)}\int \frac{G(v)}{z-v}\,\mathrm{d}v\right], \qquad \mathcal{K}(\zeta) = 1 - \frac{1 + \zeta Z(\zeta)}{\Lambda}, \tag{B2}$$

where Z is the plasma dispersion function of Fried and Conte (1961), not the charge number; we write $\mathcal{K}$ also for its continuation into the lower half plane, whose zeros are the Landau roots. On the real axis

$$\mathcal{K}(u + \mathrm{i}0) = \mathcal{K}_R(u) + \mathrm{i}\,\mathcal{K}_I(u), \qquad \mathcal{K}_R = 1 - \frac{1 + u\operatorname{Re} Z(u)}{\Lambda}, \qquad \mathcal{K}_I = -\frac{\pi\, u\, F_M(u)}{\Lambda}, \tag{B3}$$

and $\mathsf{H}[f](u) = \pi^{-1}\mathrm{P}\!\int f(v)\,(v-u)^{-1}\,\mathrm{d}v$. By the argument of Penrose (1960), $\mathcal{K}$ has no zeros in the closed upper half plane for $\Lambda \notin [0, 1]$ (section B.5). The spectral density

$$\varrho(u) = \frac{F_M(u)}{|\mathcal{K}(u + \mathrm{i}0)|^2} \tag{B4}$$

is then positive, smooth and even, with $\int \varrho\,\mathrm{d}u = 1/p_0$. For real u the equation $(L - u)F = 0$ has the Case–Van Kampen solutions

$$F_u(v) = \frac{1}{\Lambda}\,\mathrm{P}\frac{vF_M(v)}{v-u} + \mathcal{K}_R(u)\,\delta(v-u), \qquad M_0[F_u] = 1, \qquad M_1[F_u] = u, \tag{B5}$$

and the slice transform is the projection on them in the free-energy product,

$$\mathcal{T}[G](u) = [F_u, G] = \frac{\mathcal{K}_R\, G - \mathcal{K}_I\,\mathsf{H}[G]}{F_M}(u), \qquad \mathcal{T}[LG](u) = u\,\mathcal{T}[G](u), \tag{B6}$$

of which the slice amplitude (13) is the unit-weight form, $a_u = \sqrt{\varrho(u)}\,\mathcal{T}[G](u)$; $\mathcal{A}$ denotes the map $G \mapsto a$.

Theorem 4. *Let $\Lambda \notin [0, 1]$, the value $\Lambda = \infty$ included. Then L, with the domain $D(U)$ of U, is self-adjoint in $(\mathcal{W}, [\cdot,\cdot])$; it has no eigenvalues, and its spectrum is the real line, purely absolutely continuous and simple; and $\mathcal{A}$ is a unitary map from $(\mathcal{W}, [\cdot,\cdot])$ onto $L^2(\mathbb{R}, \mathrm{d}u)$ with $\mathcal{A}LG = u\,\mathcal{A}G$ for $G \in D(U)$. In particular (14) holds, $[G, \varphi(L)G] = \int \varphi(u)\, a_u^2\,\mathrm{d}u$ for every bounded Borel function φ, and*

$$G = \int \varrho(u)\,\mathcal{T}[G](u)\,F_u\,\mathrm{d}u = \mathcal{K}_R\, c + \mathcal{K}_I\,\mathsf{H}[c], \qquad c = \sqrt{\varrho}\; a. \tag{B7}$$

The rest of this appendix proves the theorem and the statements about the eigenvalues of L and the zeros of $\mathcal{K}$. For $0 < \Lambda < 1$, $\mathcal{K}$ has one zero in the upper half plane, on the imaginary axis, L has the eigenvalues $\pm\mathrm{i}\gamma$, and the slice decomposition is incomplete.

B.2 Preliminaries

We work in the complexification of $\mathcal{W}$, with $\langle f, g\rangle = \int \bar{f}\, g\, F_M^{-1}\,\mathrm{d}u$. The set $\mathcal{D}_0$ of functions pF_M, with p a polynomial, is dense in $\mathcal{W}$ (it contains the orthonormal basis e_m), lies in $D(U)$, and is mapped into itself by U, P and L. Let $\mathcal{S}$ be the set of smooth f with $|f(v)| \le C(1+|v|)^k F_M(v)$ for some C and k; it contains $\mathcal{D}_0$ and the smooth functions with compact support. For $f \in \mathcal{S}$ the Cauchy integral

$$\chi_f(\zeta) = \frac{1}{\pi}\int \frac{f(v)}{v-\zeta}\,\mathrm{d}v \qquad (\operatorname{Im}\zeta > 0)$$

is analytic in the upper half plane, extends continuously to the real axis with $\chi_f(u+\mathrm{i}0) = \mathsf{H}[f](u) + \mathrm{i}f(u)$ (the Plemelj formula), and is $O(1/\zeta)$ as $|\zeta| \to \infty$ in the closed upper half plane. The following converse is used twice.

Lemma 1. *Let Y be analytic in the upper half plane, continuous up to the real axis, and $O(1/\zeta)$ as $|\zeta| \to \infty$ in the closed upper half plane. Then $Y = \chi_y$ with $y(u) = \operatorname{Im} Y(u+\mathrm{i}0)$.*

Proof. For $\operatorname{Im}\zeta > 0$, Cauchy's formula on a large half disc gives $Y(\zeta) = (2\pi\mathrm{i})^{-1}\int Y(v+\mathrm{i}0)\,(v-\zeta)^{-1}\,\mathrm{d}v$, and Cauchy's theorem for $\overline{Y(\bar\zeta)}$, analytic in the lower half plane with the same decay, gives $\int \overline{Y(v+\mathrm{i}0)}\,(v-\zeta)^{-1}\,\mathrm{d}v = 0$. Subtracting $(2\pi\mathrm{i})^{-1}$ times the second integral from the first gives $Y = \chi_y$. □

B.3 Self-adjointness

Let $\Lambda \notin [0,1]$, so that $p_0 > 0$. Since $P = (I-\mathsf{Q}) + p_0\mathsf{Q}$, the operator $P^{1/2} = I + (\sqrt{p_0}-1)\mathsf{Q}$ is bounded and symmetric, with bounded inverse, and $[f,g] = \langle P^{1/2}f, P^{1/2}g\rangle$, so $V : G \mapsto P^{1/2}G$ is unitary from $(\mathcal{W}, [\cdot,\cdot])$ onto $(\mathcal{W}, \langle\cdot,\cdot\rangle)$. Since $V - I$ has its range in the span of $F_M \in D(U)$, V maps $D(U)$ onto itself, and by (B1)

$$VLV^{-1} = P^{1/2}UP^{1/2} = U + \mathsf{K}, \qquad \mathsf{K} = (\sqrt{p_0}-1)(\mathsf{Q}U + U\mathsf{Q}), \tag{B8}$$

where $(\sqrt{p_0}-1)^2\mathsf{Q}U\mathsf{Q}$ is absent because $\mathsf{Q}U\mathsf{Q}G = M_1[F_M]\,M_0[G]\,F_M = 0$. The operator $\mathsf{K}G = (\sqrt{p_0}-1)\big[\langle uF_M, G\rangle\,F_M + \langle F_M, G\rangle\,uF_M\big]$ is symmetric, of rank two and bounded, since $\langle F_M, F_M\rangle = 1$ and $\langle uF_M, uF_M\rangle = 1/2$. By the Kato–Rellich theorem (Kato, 1966, chapter V), $U+\mathsf{K}$ is then self-adjoint on $D(U)$ in $(\mathcal{W}, \langle\cdot,\cdot\rangle)$, so $L = V^{-1}(U+\mathsf{K})V$ is self-adjoint on $D(U)$ in $(\mathcal{W}, [\cdot,\cdot])$, which is part (i) of theorem 4.

B.4 Eigenvalues

For every $\Lambda \neq 0$, a function $G \in D(U)$ is an eigenfunction of L with eigenvalue λ if and only if

$$(u-\lambda)\,G = \frac{uF_M}{\Lambda}\,M_0[G], \qquad G \neq 0.$$

If λ is not real, $G = M_0[G]\,uF_M/[\Lambda(u-\lambda)]$, and taking M_0 gives $M_0[G]\,\mathcal{K}(\lambda) = 0$ if $\operatorname{Im}\lambda > 0$ and $M_0[G]\,\overline{\mathcal{K}(\bar\lambda)} = 0$ if $\operatorname{Im}\lambda < 0$, because the integral in (B2) at a point of the lower half plane equals $\overline{\mathcal{K}(\bar\lambda)}$ (for the same reason the resolvent for $\operatorname{Im} z < 0$ is (B2) with $\overline{\mathcal{K}(\bar z)}$ in place of $\mathcal{K}(z)$). Since $M_0[G] = 0$ would give $G = 0$, a non-real λ is an eigenvalue if and only if λ or $\bar\lambda$ is a zero of $\mathcal{K}$ in the upper half plane. For real $\lambda \neq 0$, $uF_M/(u-\lambda)$ is not square integrable near $u = \lambda$, so $M_0[G] = 0$ and $G = 0$. For $\lambda = 0$, $G = M_0[G]\,F_M/\Lambda$ requires $\Lambda = 1$.

B.5 Zeros of the dielectric function

The function $\mathcal{K}$ is analytic in the upper half plane and continuous up to the real axis, where it takes the values (B3). Since $1+\zeta Z(\zeta) = -1/(2\zeta^2) + O(\zeta^{-4})$ uniformly in $\operatorname{Im}\zeta \geq 0$, $\mathcal{K} \to 1$ as $|\zeta| \to \infty$ there, so if $\mathcal{K}$ has no real zeros, the number of its zeros in the upper half plane is the winding number about 0 of the closed curve $u \mapsto \mathcal{K}(u+\mathrm{i}0)$. A real zero requires $\mathcal{K}_I(u) = 0$, hence $u = 0$, and $\mathcal{K}_R(0) = p_0$ vanishes only at $\Lambda = 1$. For $\Lambda \neq 1$, $\mathcal{K}(u+\mathrm{i}0) = 1 - \mathcal{N}(u)/\Lambda$ with $\mathcal{N}(u) = 1 + uZ(u+\mathrm{i}0)$, and the winding number of $\mathcal{K}$ about 0 equals that of $\mathcal{N}$ about the point Λ. The curve $\mathcal{N}$ has $\operatorname{Im}\mathcal{N}(u) = \pi u F_M(u)$, $\mathcal{N}(0) = 1$ and $\mathcal{N}(u) \to 0$ as $|u| \to \infty$: it leaves 0 at $u = -\infty$, runs through the lower half plane to the point 1, reached at $u = 0$, and returns to 0 through the upper half plane, meeting the real axis only at 0 and 1. It therefore winds once around every point of $(0,1)$ and around no real point outside $[0,1]$, so

$$\#\{\zeta:\ \mathcal{K}(\zeta) = 0,\ \operatorname{Im}\zeta \geq 0\} = \begin{cases} 0, & \Lambda < 0 \text{ or } \Lambda > 1, \\ 1, & 0 < \Lambda < 1, \end{cases} \tag{B9}$$

which is the criterion of Penrose (1960) for this dielectric function. By section B.4, L has no eigenvalues for $\Lambda \notin [0,1]$.

The plasma dispersion function continues to the entire function $Z(\zeta) = \mathrm{i}\sqrt{\pi}\, e^{-\zeta^2}\operatorname{erfc}(-\mathrm{i}\zeta)$, and the zeros of the continued $\mathcal{K}$ in the lower half plane are the Landau roots. They are not eigenvalues of L, because in the lower half plane the resolvent contains $\overline{\mathcal{K}(\bar{\zeta})}$, which by (B9) has no zeros there for $\Lambda \notin [0,1]$.

B.6 Spectral measure

For $\operatorname{Im} z > 0$ and real $G \in \mathcal{W}$, let

$$C_G(z) = \int \frac{G(v)}{z-v}\,\mathrm{d}v, \qquad I_G(z) = \int \frac{G(v)^2}{(z-v)\,F_M(v)}\,\mathrm{d}v.$$

By (B2), $M_0[(z-L)^{-1}G] = C_G(z)/\mathcal{K}(z)$, and since $\int uG(u)\,(z-u)^{-1}\,\mathrm{d}u = zC_G(z) - M_0[G]$, the resolvent form is

$$R_G(z) \equiv [G, (z-L)^{-1}G] = I_G(z) - \frac{z}{\Lambda}\,\frac{C_G(z)^2}{\mathcal{K}(z)}.$$

It is the Stieltjes transform $\int (z-x)^{-1}\,\mathrm{d}\mu_G(x)$ of the spectral measure μ_G of L in the state G, which Stone's formula recovers:

$$\mu_G((u_1,u_2)) + \tfrac{1}{2}\mu_G(\{u_1\}) + \tfrac{1}{2}\mu_G(\{u_2\}) = -\frac{1}{\pi}\lim_{\eta\to 0+}\int_{u_1}^{u_2} \operatorname{Im} R_G(u+\mathrm{i}\eta)\,\mathrm{d}u. \tag{B10}$$

Let $G \in \mathcal{D}_0$ be real and $\Lambda \notin [0,1]$. Then $C_G = -\pi\chi_G$ and $I_G = -\pi\chi_{G^2/F_M}$ extend continuously to the real axis, with $C_G(u+\mathrm{i}0) = -\pi(\mathsf{H}[G] + \mathrm{i}G)(u)$ and $\operatorname{Im} I_G(u+\mathrm{i}0) = -\pi G(u)^2/F_M(u)$. By (B9), $1/\mathcal{K}$ is continuous on the closed upper half plane, so $R_G(u+\mathrm{i}\eta) \to R_G(u+\mathrm{i}0)$ as $\eta \to 0+$, uniformly for u in compact sets. With $\mathcal{K}_I = -\pi u F_M/\Lambda$, the imaginary part of the boundary value is a square:

$$-\frac{1}{\pi}\operatorname{Im} R_G(u+\mathrm{i}0) = \frac{G^2}{F_M} - \frac{\mathcal{K}_I}{F_M}\,\frac{2\mathcal{K}_R\,G\,\mathsf{H}[G] - \mathcal{K}_I(\mathsf{H}[G]^2 - G^2)}{|\mathcal{K}|^2} = \frac{\left(\mathcal{K}_R G - \mathcal{K}_I \mathsf{H}[G]\right)^2}{F_M\,|\mathcal{K}|^2} = \varrho\,\mathcal{T}[G]^2 = a_u^2.$$

By (B10) and the uniform convergence, μ_G has no atoms and $\mathrm{d}\mu_G = a_u^2\,\mathrm{d}u$.

Three consequences follow. First, the absolutely continuous subspace of L is closed and contains the dense set $\mathcal{D}_0$, so it is $\mathcal{W}$, and L has no eigenvalues and no singular continuous spectrum. Second, K in (B8) has rank two and is therefore of trace class, so by the Kato–Rosenblum theorem (Kato, 1966, chapter X) the absolutely continuous parts of $U + \mathsf{K} = VLV^{-1}$ and of U are unitarily equivalent; since both operators are purely absolutely continuous, L is unitarily equivalent to U, multiplication by u in $L^2(\mathbb{R}, \mathrm{d}u/F_M)$, whose spectrum is the real line with multiplicity one. This is part (ii) of theorem 4. Third, applying $\mathrm{d}\mu_G = a_u^2 \,\mathrm{d}u$ to $G+G'$ and $G-G'$ gives, for real $G, G' \in \mathcal{D}_0$ and every bounded Borel function φ,

$$[G', \varphi(L)G] = \int \varphi(u)\, a_u[G']\, a_u[G] \,\mathrm{d}u, \qquad a[G] = \mathcal{A}G, \tag{B11}$$

and by linearity the same holds for complex $G, G' \in \mathcal{D}_0$ with $a_u[G']$ replaced by its complex conjugate.

B.7 The slice transform

The formula (13), with H the Hilbert transform on $L^2(\mathbb{R}, \mathrm{d}u)$, defines a bounded operator $\mathcal{A}$ from $\mathcal{W}$ into $L^2(\mathbb{R}, \mathrm{d}u)$: for $\Lambda \notin [0,1]$, $|\mathcal{K}(u + \mathrm{i}0)|$ is bounded below, since it has no real zeros and tends to 1 at infinity, and since $|\mathcal{K}_R| \le |\mathcal{K}|$, the first term of (13) has norm at most $\langle G, G\rangle^{1/2}$. In the second, $\mathcal{K}_I \mathsf{H}[G]/(\sqrt{F_M}\,|\mathcal{K}|) = -\pi u \sqrt{F_M}\,\mathsf{H}[G]/(\Lambda|\mathcal{K}|)$, the factor $u\sqrt{F_M}/|\mathcal{K}|$ is bounded, H is an isometry, and $\int |G|^2 \,\mathrm{d}u \le \pi^{-1/2}\langle G, G\rangle$. By (B11) with $\varphi = 1$, $\mathcal{A}$ is isometric on the dense set $\mathcal{D}_0$, so it is an isometry of $(\mathcal{W}, [\cdot,\cdot])$ into $L^2(\mathbb{R}, \mathrm{d}u)$, and its range is closed.

The following argument is the counterpart for L of the invertibility proof of Morrison (2000). To show that the range of $\mathcal{A}$ is dense, let b be a smooth function with compact support, $c = \sqrt{\varrho}\, b$, and

$$G = \mathcal{K}_R c + \mathcal{K}_I \mathsf{H}[c] = \operatorname{Im}\bigl[\mathcal{K}(u + \mathrm{i}0)\, \chi_c(u + \mathrm{i}0)\bigr].$$

Since c has compact support, $\mathcal{K}_I = -\pi u F_M/\Lambda$ and $\mathsf{H}[c]$ is smooth and bounded, G belongs to $\mathcal{S}$ and to $\mathcal{W}$. The product $\mathcal{K}\chi_c$ is analytic in the upper half plane, continuous up to the real axis and $O(1/\zeta)$, so by lemma 1 it equals χ_G, and $\chi_c(u + \mathrm{i}0) = \chi_G(u + \mathrm{i}0)/\mathcal{K}(u + \mathrm{i}0)$. Taking imaginary parts, with the Plemelj formula for χ_G,

$$c = \frac{\operatorname{Im}\bigl[(\mathsf{H}[G] + \mathrm{i}G)\,\bar{\mathcal{K}}\bigr]}{|\mathcal{K}|^2} = \frac{\mathcal{K}_R G - \mathcal{K}_I \mathsf{H}[G]}{|\mathcal{K}|^2} = \varrho\, \mathcal{T}[G],$$

so that $\mathcal{A}G = b$. The smooth functions with compact support are dense in $L^2(\mathbb{R}, \mathrm{d}u)$ and the range of $\mathcal{A}$ is closed, so $\mathcal{A}$ is onto, with the inverse

$$\mathcal{A}^{-1} a = \mathcal{K}_R \sqrt{\varrho}\, a + \mathcal{K}_I\, \mathsf{H}\bigl[\sqrt{\varrho}\, a\bigr], \tag{B12}$$

which is (B7): since $|\mathcal{K}_R|\sqrt{\varrho/F_M} \le 1$ and $|\mathcal{K}_I|/\sqrt{F_M}$ and ϱ are bounded, the right-hand side is a bounded operator from $L^2(\mathbb{R}, \mathrm{d}u)$ into $\mathcal{W}$ that equals $\mathcal{A}^{-1}$ on a dense set. Conversely, for $G \in \mathcal{D}_0$, $Y = \chi_G/\mathcal{K}$ is analytic in the upper half plane, because $\mathcal{K}$ has no zeros there, and $\operatorname{Im} Y(u + \mathrm{i}0) = \varrho\mathcal{T}[G] = c \in \mathcal{S}$; by lemma 1, $Y = \chi_c$, and $G = \operatorname{Im}\chi_G(u+\mathrm{i}0) = \operatorname{Im}[\mathcal{K}\,Y](u+\mathrm{i}0) = \mathcal{K}_R c + \mathcal{K}_I \mathsf{H}[c]$. The first direction does not use the zeros of $\mathcal{K}$ in the upper half plane; the second uses their absence, and fails by the residue of Y at the zero $\mathrm{i}\gamma$ that $\mathcal{K}$ has for $0 < \Lambda < 1$, where $1 - \sqrt{\pi}\,\gamma\, e^{\gamma^2} \operatorname{erfc}(\gamma) = \Lambda$ and L has the eigenvalues $\pm\mathrm{i}\gamma$ (section B.4); the slices are then not complete. The KRMHD branches never fall in this range, because $\Lambda^+ > 1$ and $\Lambda^- < 0$.

Part (iii) of theorem 4 follows. By (B11) with $\varphi(u) = (z-u)^{-1}$, $\operatorname{Im} z \ne 0$, $[G', (z-L)^{-1}G] = \int \overline{a_u[G']}\, a_u[G]\, (z-u)^{-1} \,\mathrm{d}u$ for $G, G' \in \mathcal{D}_0$, and by continuity for all $G, G' \in \mathcal{W}$. Since $\mathcal{A}$ is unitary,

the left-hand side is $\int \overline{a_u[G']}\, a_u[(z-L)^{-1}G]\,\mathrm{d}u$, and since $\mathcal{A}G'$ runs over $L^2(\mathbb{R},\mathrm{d}u)$, $\mathcal{A}(z-L)^{-1}G = (z-u)^{-1}\mathcal{A}G$. For $G \in D(U)$, applying this to $(z-L)G$ shows that $u\,\mathcal{A}G \in L^2(\mathbb{R},\mathrm{d}u)$ and $\mathcal{A}LG = u\,\mathcal{A}G$. The relations (14) are then the functional calculus of multiplication by u.

By (B5), $\int c(u)\,F_u(v)\,\mathrm{d}u = \mathcal{K}_R(v)\,c(v) + \mathcal{K}_I(v)\,\mathsf{H}[c](v)$, so (B12) is the completeness relation $G = \int \varrho\,\mathcal{T}[G]\,F_u\,\mathrm{d}u$ of the modes, and $\varrho\,\mathcal{T}[\mathcal{A}^{-1}a] = \sqrt{\varrho}\,a$ is the orthogonality relation $[F_u, F_{u'}] = \delta(u-u')/\varrho(u)$, in the sense of distributions.

C The truncated hierarchy

C.1 Weights, closures and eigenvectors

Let J be the ladder matrix of the zero closure at $\Lambda = \infty$, the symmetric Jacobi matrix of the Hermite polynomials h_m orthonormal with weight F_M, and J_c that of the copy closure, which for $M \geq 2$ differs from J only in $(J_c)_{M,M-1} = \sqrt{M/2} + \sqrt{(M+1)/2}$. A finite Λ changes only $A_{10} = (1-1/\Lambda)\,J_{10}$, and column 0 of J and of J_c has the single non-zero entry $J_{10} = 1/\sqrt{2}$. Hence, with $P_\Lambda = \mathrm{diag}(1-1/\Lambda, 1, \ldots, 1)$,

$$A = J\,P_\Lambda \quad \text{(zero closure)}, \qquad A = J_c\,P_\Lambda \quad \text{(copy closure, } M \geq 2\text{)}, \tag{C1}$$

the Hermite form of $L = UP$ (B1). For the zero closure $\mathsf{P} = P_\Lambda$, and $\mathsf{P}A = P_\Lambda J P_\Lambda$ is symmetric. For the copy closure $D_c = \mathrm{diag}(1, \ldots, 1, p_M)$ satisfies $D_c J_c = J$, because $p_M (J_c)_{M,M-1} = \sqrt{M/2}$, so that with $\mathsf{P} = D_c P_\Lambda$ again $\mathsf{P}A = P_\Lambda J P_\Lambda$.

Since $LF_u = uF_u$ and the untruncated ladder matrix is the matrix of L in the Hermite basis, the Hermite coefficients $r_m(u)$ of F_u satisfy $\sum_n A_{mn} r_n = u\,r_m$ row by row, with $r_0 = M_0[F_u] = 1$:

$$r_1 = \sqrt{2}\,u, \qquad r_2 = u\,r_1 - \frac{1-1/\Lambda}{\sqrt{2}}\,r_0, \qquad \sqrt{\frac{m+1}{2}}\,r_{m+1} = u\,r_m - \sqrt{\frac{m}{2}}\,r_{m-1} \quad (m \geq 2). \tag{C2}$$

Thus r_m is a polynomial of degree m with $r_m(-u) = (-1)^m r_m(u)$, and $r_m = h_m$ at $\Lambda = \infty$. Rows 0 to $M-1$ of the truncated matrix coincide with those of the infinite matrix for both closures, and row M is the condition on the eigenvalues,

$$r_{M+1}(u_j) = 0 \quad \text{(zero closure)}, \qquad r_{M+1}(u_j) = r_{M-1}(u_j) \quad \text{(copy closure)}, \tag{C3}$$

the second from row M of the copy closure, $u\,r_M = [\sqrt{M/2} + \sqrt{(M+1)/2}\,]\,r_{M-1}$, and the recurrence at $m = M$.

Proof of theorem 2. (i) A is similar to the symmetric tridiagonal matrix $\mathsf{P}^{1/2} A\,\mathsf{P}^{-1/2}$, whose off-diagonal entries are non-zero because $A_{01}A_{10} = (1-1/\Lambda)/2 > 0$, so its eigenvalues are real and simple and the vectors $\mathsf{P}^{1/2}\boldsymbol{r}_j$ are orthogonal. By (C2) the vector $(r_0(u), \ldots, r_M(u))^{\mathsf{T}}$ satisfies rows 0 to $M-1$ of $A\boldsymbol{r} = u\boldsymbol{r}$ for every u, (C3) fixes the u_j, and $\mathsf{P}A = A^{\mathsf{T}}\mathsf{P}$ gives $\boldsymbol{w}_j^{\mathsf{T}} A = u_j \boldsymbol{w}_j^{\mathsf{T}}$. (ii) The combination $\boldsymbol{w}_j^{\mathsf{T}} g$ commutes with the brackets, with ∂_z and with the linear operations, and $\boldsymbol{w}_j^{\mathsf{T}} A = u_j \boldsymbol{w}_j^{\mathsf{T}}$ turns $s\nabla_{\parallel}(Ag)$ into $s u_j \nabla_{\parallel}(\boldsymbol{w}_j^{\mathsf{T}} g)$; the converse follows from (iii). (iii) By (i), $R^{\mathsf{T}}\mathsf{P}R = I$, so $g = Ra$ and $\sum_m p_m |g_m|^2 = a^\dagger R^{\mathsf{T}}\mathsf{P}R\,a = \sum_j |a_j|^2$. □

C.2 Gauss quadrature

We prove proposition 2. For the basis functions e_m, $\mathcal{T}[e_m](u) = [F_u, e_m] = p_m\,r_m(u)$. (i) By theorem 4 the slice transform is unitary, and the slice amplitude of e_m is $\sqrt{\varrho}\,p_m r_m$, so $p_m \delta_{mn} = [e_m, e_n] = \int \varrho\,p_m r_m\,p_n r_n\,\mathrm{d}u$. (ii) The polynomials $\sqrt{p_m}\,r_m$ are therefore orthonormal for $\varrho\,\mathrm{d}u$, with

the Jacobi matrix $\mathsf{P}^{1/2} A\, \mathsf{P}^{-1/2}$ by (C2), and the zero closure keeps its first $M+1$ rows and columns. By the theorem of Golub and Welsch (1969), the eigenvalues of this truncated matrix are the zeros of r_{M+1} and the nodes of the Gauss rule, and the weights are $\mu_0 O_{0j}^2$, with O its orthonormal eigenvectors and $\mu_0 = \int \varrho \, \mathrm{d}u = 1/p_0$. Since $O_{mj} = \sqrt{p_m}\, R_{mj}$, the weights are $R_{0j}^2 = 1/n_j$, and exactness to degree $2M+1$ is Gauss's theorem. (iii) By definition $\boldsymbol{r}_j$ holds the Hermite coefficients of F_{u_j}, and $\mathcal{T}[G](u_j) = \sum_m g_m p_m r_m(u_j) = \boldsymbol{w}_j^{\mathsf{T}} g = \sqrt{n_j}\, a_j$.

C.3 The largest phase velocity

For the zero closure $A = J + \delta A$ exactly, with $\delta A = -(\Lambda\sqrt{2})^{-1} \boldsymbol{e}_1 \boldsymbol{e}_0^{\mathsf{T}}$ and $\boldsymbol{e}_m$ the mth unit vector. The eigenvalues of J are the zeros x_j of h_{M+1}, and its normalised eigenvectors have components $(\lambda_j^{\infty})^{1/2} h_m(x_j)$, with λ_j^{∞} the Gauss–Hermite weights. Since $h_0 = 1$ and $h_1(x) = \sqrt{2}\, x$, to first order $u_j = x_j - \lambda_j^{\infty} x_j/\Lambda + O(\Lambda^{-2})$. The weight λ_M^{∞} is of order $e^{-x_M^2}$ up to an algebraic factor, with $x_M^2 = 2M + 3 - 2^{2/3}|\alpha_1|(2M+3)^{1/3} + \ldots$, so the shift of $u_{\max}$ is about 10^{-49} at $M = 64$. Up to such terms, $u_{\max}$ is the largest zero of ψ_{M+1} for the zero closure and the largest root of $\psi_{M+1} = \psi_{M-1}$ for the copy closure.

The function ψ_{M+1} solves $\psi'' + (q^2 - x^2)\psi = 0$ with $q = q_{M+1}$. With $x = q + (2q)^{-1/3}\xi$ and $\sigma = (2q)^{-4/3}$, the uniform Airy (Langer) approximation $\psi \propto (\chi')^{-1/2} \mathrm{Ai}(\chi)$, with $\chi = \xi + \sigma\xi^2/5 + O(\sigma^2)$, places the largest zero at $\chi = \alpha_1$, which gives the first line of (28), the classical expansion of Szegő (1975, theorem 6.32) and Olver (1959) (DLMF, 2026, eq. 12.11.9). For the copy closure, $x\psi_M = \sqrt{M/2}\,\psi_{M-1} + \sqrt{(M+1)/2}\,\psi_{M+1}$ and $\psi'_M = \sqrt{M/2}\,\psi_{M-1} - \sqrt{(M+1)/2}\,\psi_{M+1}$ turn row M, $\psi_{M+1} = \psi_{M-1}$, into the Robin condition $\psi'_M + x(\sqrt{M} + \sqrt{M+1})^{-2}\psi_M = 0$ on ψ_M, which solves the same equation with $q = q_M$ and to leading order reads $\psi_\xi + \sigma\psi = 0$. The same approximation turns this condition into

$$\mathrm{Ai}'(\chi) + \tfrac{4}{5}\,\sigma\,\mathrm{Ai}(\chi) = 0, \qquad \text{so that} \qquad \chi = \alpha_1' - \frac{4\sigma}{5\alpha_1'} + O(\sigma^2),$$

and $\xi = \chi - \sigma\chi^2/5$ gives the second line of (28); $u_{\max}$ up to $M = 16384$ confirms both lines.

D Numerical methods

D.1 Runs and checks

The time step is GANDALF's Lawson–Runge–Kutta scheme (Lawson-RK4; Lawson, 1967), which integrates the streaming exactly through an integrating factor and treats the nonlinear terms with explicit RK4. Following section 4.4, $\Delta t = 0.8 \times 2\sqrt{2}/\rho$, where ρ is the spectral radius of the explicit slice-advection operator at the extreme nodes $u = \pm u_{\max}$, computed by a Lanczos iteration every 25 steps, or the Alfvénic limit $0.8\,\Delta x / \max|\nabla_\perp z^{\pm}|$ if that is smaller. The runs at $\beta_i = 0.1$ and at $M \geq 64$ use a safety factor of 0.4 on the Alfvénic limit, with both limits updated every step. The runs at $M = 32$ used GANDALF version 0.7.0 on a CPU, and the runs at $M \geq 64$ versions 0.8.0 and 0.8.1, whose solver is the same, on a GPU.

The Alfvénic turbulence is first spun up with the same forcing and dissipation. Each run then starts from $g = 0$, and the averages are over the windows of tables 2 and 3. At $\beta_i = 1$, 64^3 and $M = 32$ the run lasts $53\,\tau_A$ and the window is the final $50\,\tau_A$; W reaches its mean level within about $3\,\tau_A$. The window starts at $t = 15.5$ in the run at $\beta_i = 0.3$ and $M = 32$ and at $28.5\,\tau_A$ in the run at $\beta_i = 0.3$ and $M = 192$, because the free energy was still settling before. At $\beta_i = 0.05$ the free energy fluctuates slowly, and those runs are averaged over 160–300 τ_A.

Table 4: Checks of the run at $\beta_i = 1$, 64^3 and $M = 32$ over the window $3 \le t \le 53$.

Check	Alfvénic	Compressive
Time-stepping error / injected energy	5.0×10^{-4}	3.8×10^{-8}
|injection − dissipation| / injection	1.7%	0.06%
Standard deviation / mean of E_A and W	0.117	0.141
Spread of block means, four blocks of $12.5\,\tau_A$	2.4% (E_A)	8.8% ($W(m)$), 6.9% (Π), 15.5% ($D_\perp$), 8.4% (D_ν)
Energy fraction beyond $k_\perp/k_1 = 21$	4×10^{-5}	2×10^{-5}
Free energy at $m > 3M/4$	–	1.1%
Share of D_ν at $m \ge M/2$	–	95%

Table 4 lists the checks of the run at $\beta_i = 1$, 64^3 and $M = 32$, figure 10 shows that run, and table 5 gives the fluctuations and convergence of every run. Energy is conserved by the time step and balanced between injection and dissipation, in every run to within 2.1×10^{-3} of the injection and to within 2%. The Alfvénic energy fluctuates by 8–16% about its mean, which at this box size is intrinsic to the forced turbulence. The free energy fluctuates more slowly as β_i decreases, so its averages are less certain at low β_i. We quote the spread of the four block means of each average, each divided by the injection rate of its block, as its uncertainty: at $\beta_i = 1$ and $M = 32$ the block means agree to within 9% except $D_\perp$ at $m = 0$ (15.5%), and at $\beta_i = 0.1$ the spread is about ±12% on $W(m)$. At most 2.2% of the free energy lies at $m > 3M/4$ in any run of table 2, and the spectra show no sign of numerical instability at small scales in any run.

D.2 Energy accounting

Each step of the simulations of section 7 consists of GANDALF's Lawson-RK4 step with dissipation and hypercollisions, followed by the forcing increment. The hyper-resistivity and the hypercollisions are linear and diagonal in Fourier and Hermite space, so the free energy they remove from each moment during a step follows from the post-step fields and their damping factors, and the energy before dissipation is reconstructed exactly. The difference between that energy and the energy at the start of the step is the change due to the ideal part of the step, which the exact dynamics conserves; summed over the window it is the time-stepping error of table 4. The injected energy of each step is the change across the forcing increment. This gives $D_\perp(m)$, $D_\nu(m)$ and the injection rate in (21) per moment without finite differences in time, and the field-aligned flux is computed from (20) with GANDALF's dealiased bracket.

D.3 Field-line tracing

Field lines of $\hat{\boldsymbol{b}} = \hat{\boldsymbol{z}} + \hat{\boldsymbol{z}} \times \nabla_\perp \Psi$ are integrated in z with the classical fourth-order Runge–Kutta method and step $\Delta z = L/128$ ($L/256$ at 256^3), with $\nabla_\perp \Psi$ interpolated by cubic B-splines on a grid refined by a factor of two by zero padding. We start 4096 lines at random points of the plane $z = 0$ and follow each over four box lengths. Along each line we sample the field of interest and compute its spectrum in the arc length, which equals z to first order in $b_\perp$. The mean field-aligned wavenumber κ_1 is the first moment of $|\kappa|$ of the spectrum averaged over lines, and we average it over the snapshots of the run. For the injected pattern we draw four fresh realisations of the density forcing with the run's generator and trace them on the same lines; their field-aligned spectrum, averaged over lines, draws and snapshots, is $P(\kappa)$.

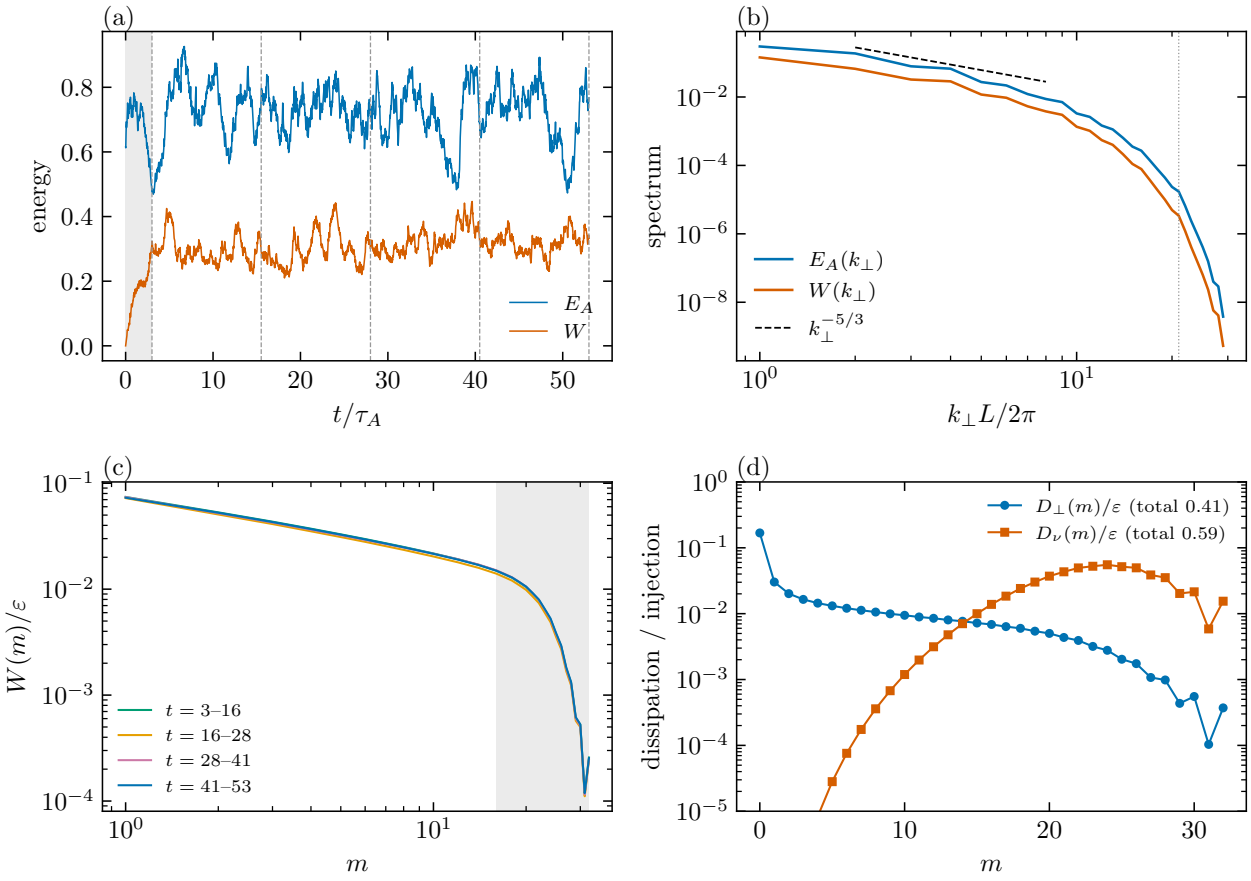


Figure 10: Overview of the run at $\beta_i = 1$, 64^3, $M = 32$. (a) E_A and W against t; the dashed lines bound the four averaging blocks, and the shading marks the excluded start. (b) $E_A(k_\perp)$ and $W(k_\perp)$ averaged over $3 \le t \le 53$, with $k_\perp^{-5/3}$ for reference; the dotted line is the edge of the dealiasing band. (c) $W(m)/\varepsilon$ in each block, divided by the injection rate of that block; the shading marks $m > M/2$, where the hypercollisions act. (d) The rates at which the hyper-resistivity, $D_\perp(m)$, and the hypercollisions, $D_\nu(m)$, remove free energy from moment m, as fractions of the injection rate; in total they remove 41% and 59% of it.

D.4 The frozen copy

To separate structure of g_0 along the field from the effect of filtering it in $k_\perp$, we build from each snapshot a field f that equals g_0 on the planes $z_n = nL/4$ and, between two neighbouring planes $z_n \le z < z_{n+1}$, is the blend

$$f(\boldsymbol{x}_\perp, z) = \cos^2\Big(\frac{\pi(z - z_n)}{2a}\Big)\, g_0(\boldsymbol{X}_n, z_n) + \sin^2\Big(\frac{\pi(z - z_n)}{2a}\Big)\, g_0(\boldsymbol{X}_{n+1}, z_{n+1}), \qquad a = \frac{L}{4}, \tag{D1}$$

where $\boldsymbol{X}_n$ and $\boldsymbol{X}_{n+1}$ are the points at which the field line through $(\boldsymbol{x}_\perp, z)$ crosses the two planes, traced as above from every grid point. Along any field line f is a smooth interpolation of the values of g_0 at the plane crossings, so it has no field-aligned structure on scales shorter than a apart from that of the blend, while across the field it has the structure of g_0; its variance is 83–84% of that of g_0. We trace the same lines as for g_0 on the same snapshots, filter f, g_0 and Φ into the same bands of $k_\perp$, and compute their spectra along the lines.

Table 5: Fluctuations and convergence of the runs of table 2 (upper part, in the same order) and of table 3 (lower part). std/mean is that of E_A and W over the window; the block spread is the largest relative spread of the four block means of $W(m)$, Π, $D_\perp(m)$ and $D_\nu(m)$; $W_>$ is the fraction of the free energy at $m > 3M/4$.

β_i	N	M	std/mean (E_A, W)	block spread	$W_>$
1	64	32	0.117, 0.141	0.155	1.1%
0.3	64	32	0.076, 0.147	0.174	1.0%
0.1	64	32	0.120, 0.180	0.229	0.65%
1	64	64	0.146, 0.143	0.312[a]	1.2%
0.3	64	64	0.158, 0.136	0.185	0.87%
0.1	64	64	0.114, 0.186	0.212	0.44%
0.3	96	64	0.133, 0.153	0.238[b]	1.02%
0.3	128	64	0.114, 0.156	0.236[a]	1.06%
0.3	256	64	0.112, 0.123	0.162	1.19%
1	128	64	0.112, 0.165	0.143	1.35%
1[c]	64	64	0.156, 0.080	0.321[a]	2.15%
β_i, χ	N	M	std/mean (E_A, W)	block spread	
0.05, 0.96	64	128	0.117, 0.172	0.47[d]	
0.1, 0.96	64	64	0.114, 0.186	0.21	
0.3, 0.94	64	192	0.124, 0.095	0.26[e]	
0.05, 0.75	64	128	0.135, 0.139	0.18	
0.05, 0.56	64	256	0.150, 0.143	0.12	
0.05, 0.40	64	512	0.102, 0.107	0.16	

[a] $D_\perp(m)$ at the forced moment: at $m = 0$ (31% and 24%; 20% at $m = 1$ for $\beta_i = 1$), and at $m = 1$ in the run with g_1 forced (32%, and 29% for $W(1)$); at most 19% for every other average and m. [b] $W(m)$ at $m = 47$ and $D_\nu(m)$ at $m = 51$, in the collisional range. [c] g_1 forced in place of g_0. [d] $W(m)$ and $D_\nu(m)$ at $m = 65$, in the collisional range, from the slow fluctuation of W; on the fit range the four block fits of ℓ_* agree to within 15%. [e] $D_\perp$ at $m = 0$; every other average within 19%.

D.5 The two-channel model

For one field-aligned wavenumber κ the model is the forced linear hierarchy

$$\frac{\mathrm{d}g}{\mathrm{d}t} = -\mathsf{L}g + \xi\,\boldsymbol{e}_0, \qquad \mathsf{L} = \mathrm{i}s\kappa A + \mathrm{diag}\Big(\frac{1}{2T} + \nu\Big(\frac{m}{M}\Big)^n[m \geq 2]\Big), \tag{D2}$$

where A is the truncated ladder matrix of the run, $\boldsymbol{e}_0$ the unit vector of g_0 and ξ white noise of intensity q. The damping $1/(2T)$ of every amplitude removes free energy from every moment at the rate $1/T$. The stationary covariance $\mathsf{C} = \langle gg^\dagger\rangle$ solves the Lyapunov equation $\mathsf{LC} + \mathsf{CL}^\dagger = q\,\boldsymbol{e}_0\boldsymbol{e}_0^\mathsf{T}$. Then $W(m) = p_m\mathsf{C}_{mm}$, the flux is $\Pi_{m+1/2} = -2s\kappa\,p_m A_{m,m+1}\,\mathrm{Im}\,\mathsf{C}_{m+1,m}$, and the injection rate is $p_0 q$. The model spectrum is the sum of these over the measured $P(\kappa)$, normalised so that the total injection rate equals the measured ε. Free energy injected at $\kappa = 0$ does not phase mix and is removed at $m = 0$ at the rate $1/T$. At large m and without hypercollisions, the sum reproduces (44).

D.6 Code and data

The configuration files, the run script and the analysis scripts are in the public repository https://github.com/anjor/krmhd-research, in the folders `configs/`, `scripts/` and `analysis/` of `paper/phase-velocity-slices/`; the analysis scripts read only saved data and regenerate every number and figure of section 7 and of this appendix. The self-checking scripts behind sections 3 to 5 are in `derivations/` of the same folder. The time series, snapshots and analysis outputs of the runs are available from the author on request.

E Notation

Table 6 compares our notation with that of CMM (Chandran et al., 2026), S09 (Schekochihin et al., 2009) and the GANDALF code (Kanekar, 2025). Three symbols mean different things in the three papers. Our κ is the wavenumber of the compressive fluctuations along the perturbed field, which S09 (§6.3) call $k_{\parallel 0}$; CMM use κ for the square root that we call Δ. Our χ is the nonlinearity parameter, whereas in S09 and CMM χ is the gyrokinetic potential. Our $W(m)$ is the Hermite spectrum of one branch, whereas in S09 and CMM W is the free energy of the whole plasma, which S09 call the generalized energy. S09 also use ζ twice, for the Elsasser potentials $\zeta^{\pm}$ and for the phase velocity ζ_i of a linear mode.

Table 6: Notation of this paper, CMM (Chandran et al., 2026), S09 (Schekochihin et al., 2009) and GANDALF (Kanekar, 2025), with equation or section numbers of the respective papers; "same" means as in this paper.

This paper	CMM	S09	GANDALF
Parameters and velocity			
$v_{\rm th} = (2T_i/m_i)^{1/2}$, $s = v_{\rm th}/v_A = \beta_i^{1/2}$	same	same	$s = \beta_i^{1/2}$
$u = v_\parallel/v_{\rm th}$, also the phase velocity of a slice; $F_M(u) = \pi^{-1/2}e^{-u^2}$	$\hat{v}_\parallel$; F_M in $v_\parallel$ (below B4)	$\zeta_i = \omega/\lvert k_\parallel\rvert v_{\rm th}$ for a linear mode (§6.2.2); F_M as CMM	same
$\Lambda^\pm$ of (6); $\Lambda = \infty$ for $l \geq 2$	same (B7)	same (182)	input
Δ in (6)	κ (B16, printed without the square)	written out (182)	—
$\chi = k_\perp u_\perp/(k_\parallel v_A)$, the nonlinearity parameter	$\chi = \phi - \boldsymbol{v}\cdot\boldsymbol{A}/c$, the gyrokinetic potential (2.7)	χ as CMM; critical balance is our $\chi \sim 1$	—
Alfvénic fields			
Φ, with $\boldsymbol{u}_E = \hat{\boldsymbol{z}}\times\nabla_\perp\Phi$	$\varphi = 2\Phi/(\rho_i v_{\rm th})$ (3.2); $\boldsymbol{u}_\perp = \boldsymbol{u}_E$	$\Phi = c\phi/B_0$ (135); $\boldsymbol{u}_\perp = \boldsymbol{u}_E$	$\Phi = (z^+ + z^-)/2$
$\Psi = -A_\parallel/(4\pi n_i m_i)^{1/2}$	$A = -\Psi/(\rho_i v_A)$ (3.2)	same (135)	$\Psi = (z^+ - z^-)/2$
$\nabla_\parallel = \hat{\boldsymbol{b}}\cdot\nabla = \partial_z + \{\Psi,\cdot\}$, with $\boldsymbol{b}_\perp = \hat{\boldsymbol{z}}\times\nabla_\perp\Psi$	$\partial_z + \boldsymbol{b}_\perp\cdot\nabla_\perp/v_A = \partial_z - \rho_i\{A,\cdot\}$ (3.4)	$\hat{\boldsymbol{b}}\cdot\nabla$	same
$z^\pm = \Phi \pm \Psi$; z^+ propagates toward $-z$	same by (3.23)–(3.24)	$\zeta^\pm$ (21)	same
κ (sections 6 and 7) and $k_\parallel$ (section 5): wavenumber along the perturbed field; k_z: along $\hat{\boldsymbol{z}}$	—	$k_{\parallel 0}$ for the compressive fluctuations, $k_{\parallel A}$ for the Alfvén waves (§6.3)	k_z only
Compressive fields			
g	g_{09} (3.26), not their g_i	same (157)	—
$G^\pm$; G_l for $l \geq 2$	same (B5, B8)	same (183–184, 205)	one branch, set by Λ
$g_m = (2^m m!)^{-1/2}\int H_m(u)\,G\,{\rm d}u$	$\tilde{G}_m^\pm$ (B27, B30)	—	same
Free energy and invariants of one branch			
$W(m) = p_m\langle g_m^2\rangle$, the Hermite spectrum; $\sum_m W(m) = \int \mathcal{E}(u)\,{\rm d}u$	$\sum_m W(m) = 2W^\pm_{\rm compr}/(n_i T_i)$ (B13, B31); W (2.13) is the gyrokinetic free energy	W is the generalized energy (§3.4)	—
$\int u\,\mathcal{E}(u)\,{\rm d}u$	$2\Gamma^\pm/v_{\rm th}$ (B25, B32)	—	—
$\mathcal{H}^\pm \propto \int \mathcal{E}(u)\,{\rm d}u/u$	$H^\pm_{\text{ph-sp}}$ (B18); $\mathcal{H}$ (2.15) is the gyrokinetic helicity	—	—

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